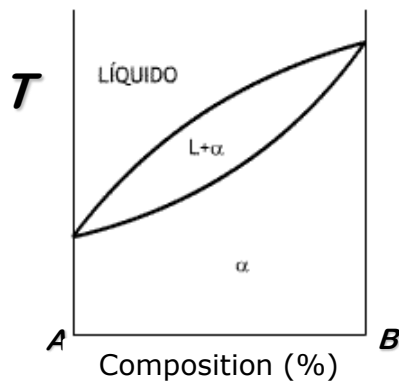


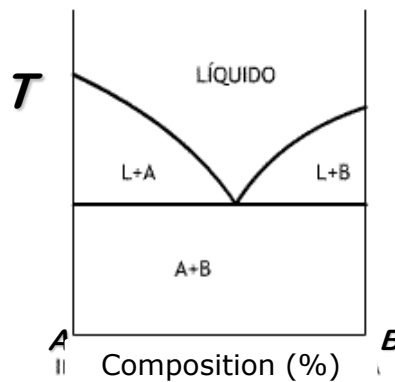
Binary phase diagrams

The solubility of two metals is defined by the Hume Rothery rules

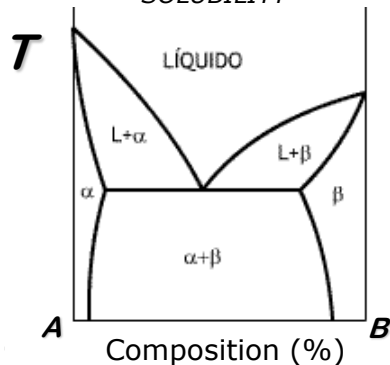
COMPLETE SOLID SOLUBILITY



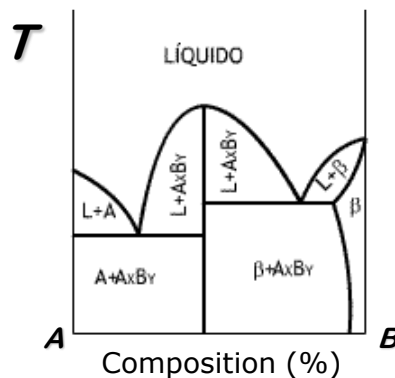
NO SOLID INSOLUBILITY



PARCIAL SOLID SOLUBILITY



INTERMEDIATE PHASES

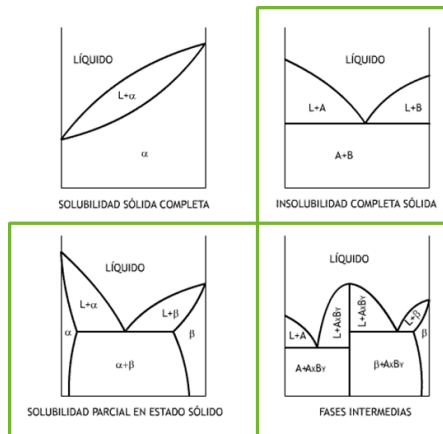


A and B Metals

Alloy.- metallic material composed of one or more metals (M o NM)

Binary Alloys

Invariant Reactions



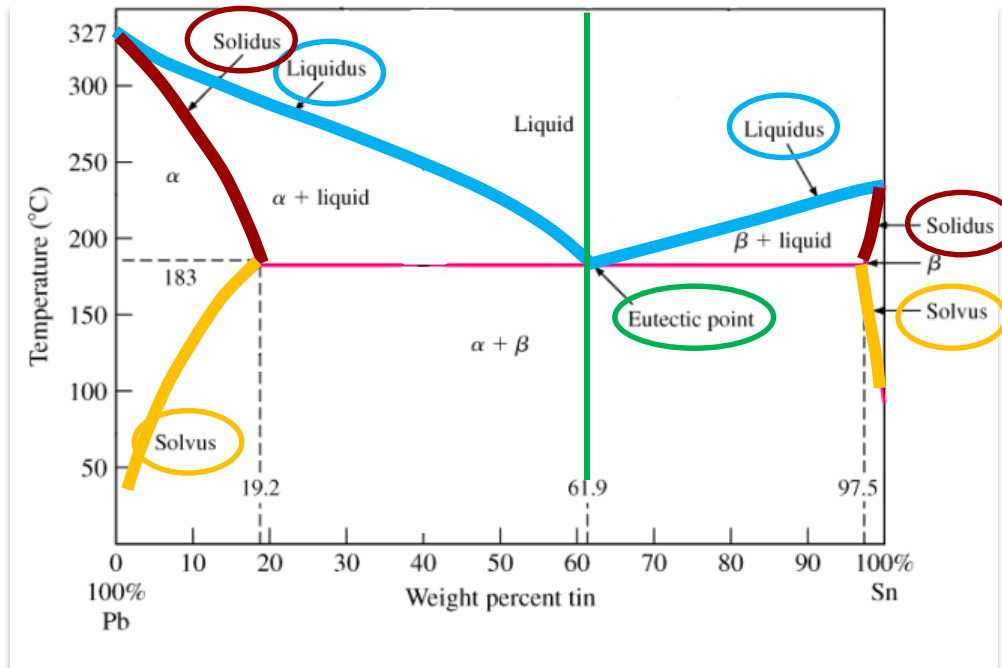
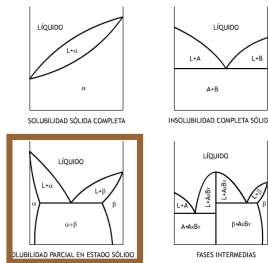
Liquid-Solid Transformations

Eutectic	$L \rightarrow \alpha + \beta$	
Peritectic	$\alpha + L \rightarrow \beta$	
Monotectic	$L_1 \rightarrow L_2 + \alpha$	
Eutectoid	$\gamma \rightarrow \alpha + \beta$	
Peritectoid	$\alpha + \beta \rightarrow \gamma$	

Solid-Solid Transformations

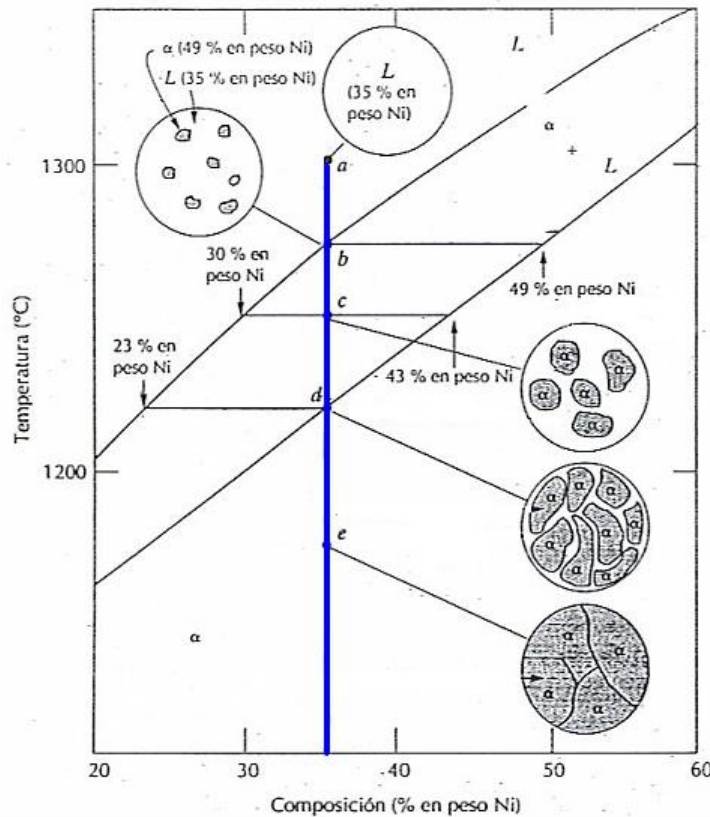
Binary systems with partial solid solubility

Partial solid solubility



Microstructure evolution

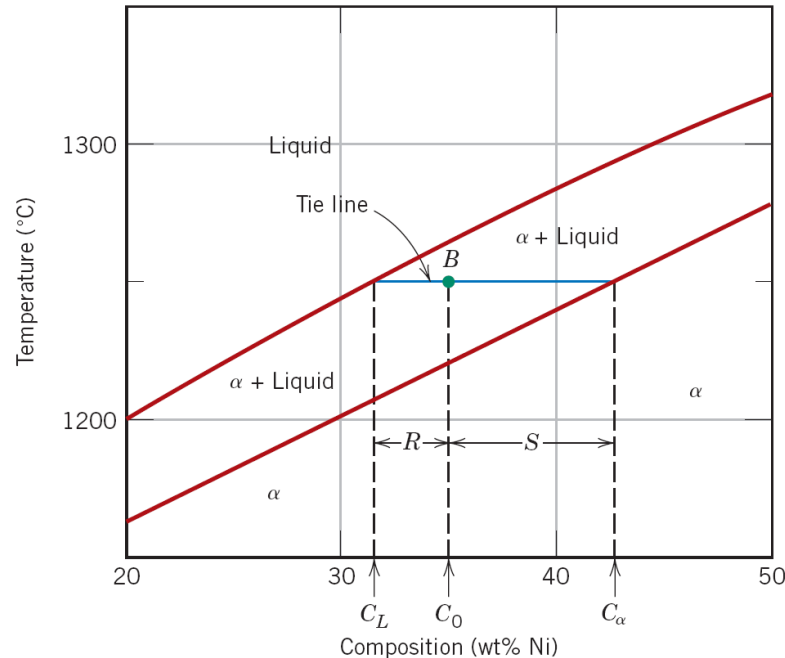
Alloy 35Ni.65Cu cooled from 1300°C.



Slow cooling so that phases remain in equilibrium.

- a) Complete liquid 35% Ni, 65% Cu.
- b) 1270°C, appear first Solid: 49% Ni, 51% Cu; Liquid: 35% Ni, 65% Cu
- c) 1250°C, Líquid: 30% Ni, 70% Cu; Solid: 49% Ni, 57% Cu
- d) 1220°C, Líquid: 23% Ni, 77% Cu; Solid: 43% Ni, 57% Cu.
- e) Solid Solution with uniform composition: 35% Ni + 65% Cu
- f) Until room temperature neither microstructure nor composition changes

Binary systems: Lever rule



$C_0 = 35\% \text{Ni}$

$C_L: 32\% \text{Ni}, C_S: 43\% \text{Ni}$

At 1100°C, alloy 35Ni.65Cu

Weight Fraction L

$$W_L = \frac{S}{R+S} = \frac{C_\alpha - C_o}{C_\alpha - C_L} = \frac{43-35}{43-32} = 0.73$$

Weight fraction of the solid phase

$$W_\alpha = \frac{R}{R+S} = \frac{C_o - C_L}{C_\alpha - C_L} = \frac{35-32}{43-32} = 0.27$$

B: 73 % L + 27 % α

$$0.73 \cdot 0.32 + 0.27 \cdot 0.43 = 0.35$$

$$W_L \cdot C_L + W_\alpha \cdot C_\alpha = C_o$$

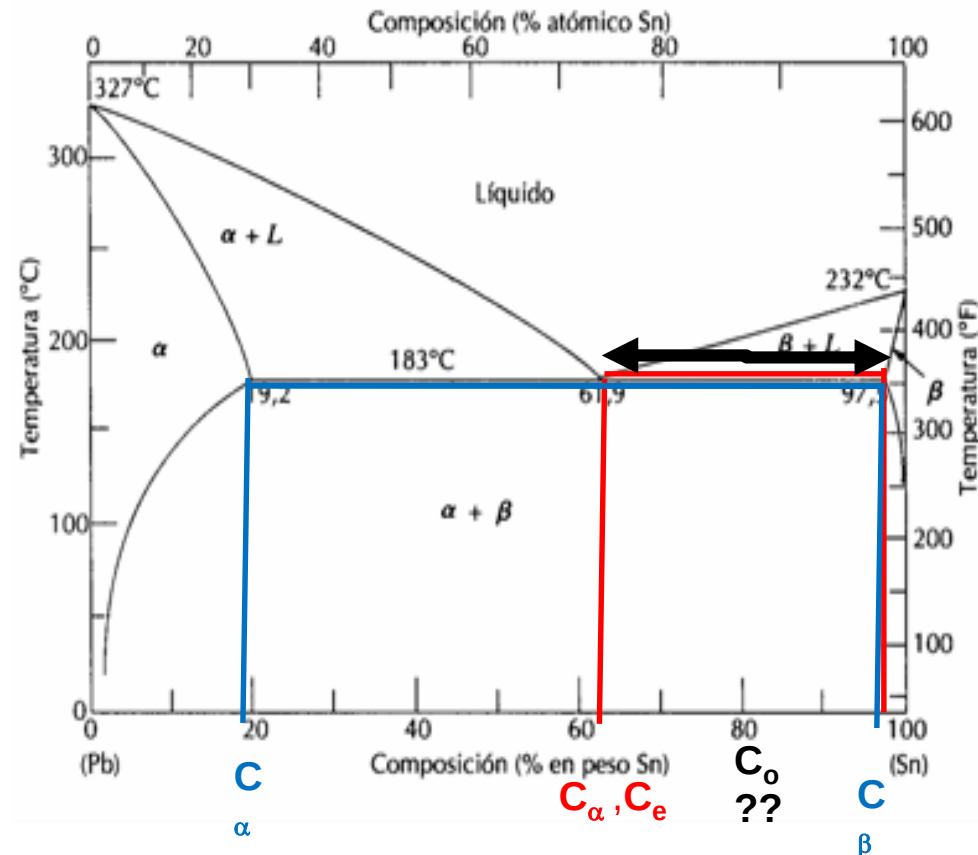
2.- For a total of 2,8 Kg of Sn-Pb alloy, is it possible to have 2,21 Kg of primary β and 2,53 Kg of total β at $183 - \Delta T$ °C?. Justify your answer.

Since we have primary β , the alloy must be **HYPEREUTECTIC**

$$\chi_{\beta'primary} = \frac{2.21Kg}{2.8Kg} = 0.79$$

$$\chi_{\beta'primary} = \frac{C_0 - C_e}{C_{\beta} - C_e} = \frac{C_0 - 61.9}{97.5 - 61.9} = 0.79$$

$$C_0 = 90\%$$



3- Using the Pb-Sn phase diagram for an alloy composition of Pb-80%Sn, indicate at the temperatures

of $T_1 = 250^\circ\text{C}$, $T_2 = 183 + \Delta T$ °C, $T_3 = 183 - \Delta T$ °C, $T_4 = 100^\circ\text{C}$

a) The phases present, chemical composition of the phases and mass fraction.

b) The micro constituents, chemical composition and mass fractions .

•Note the agreement that exists between the data obtained in (a) and (b).

a) T1; phases L; $C_L = 80\%\text{Sn}$

T2; phases α and L; $C_L = 61.9\%\text{Sn}$; $C_\beta = 97.5\%\text{Sn}$;

$$\chi_\beta = \frac{C_o - C_\beta}{C_\beta - C_\alpha} = \frac{80 - 61.9}{97.5 - 61.9} = 0.51$$

$$\chi_L = 0.49$$

(Note: this β phase is β_{primary})

T3; phases α and β ; $C_\alpha = 19.2\%$; $C_\beta = 97.5\%$

$$\chi_\alpha = \frac{C_\beta - C_o}{C_\beta - C_\alpha} = \frac{97.5 - 80}{97.5 - 61.9} = 0.22$$

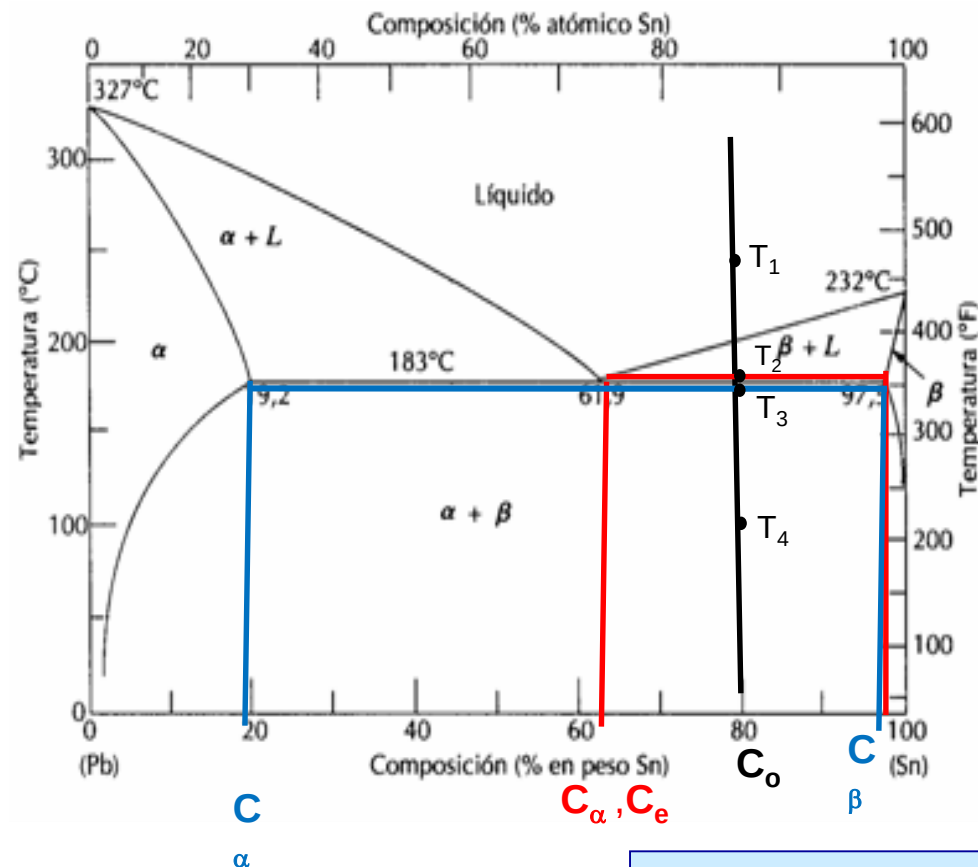
(Note: this α phase is α_{eutectic})

$$\chi_{\beta\text{TOTAL}} = 0.78$$

$$\chi_{\beta\text{TOTAL}} = \chi_{\beta\text{eutectic}} + \chi_{\beta\text{primary}}$$

$$\chi_{\beta\text{primary}} = 0.51$$

$$\chi_{\beta\text{eutectic}} = 0.27$$



T4; phases α and β ; $C_\alpha = 5\% \text{Sn}$ $C_\beta = 100\% \text{Sn}$

$$\chi_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{100 - 80}{100 - 5} = 0.21 \quad \chi_{\beta \text{TOTAL}} = 0.79$$

b) T1, T2 = microconst. are the same as phases

T3: $\beta_{\text{primary}} + \text{eutectic}(\alpha + \beta)$

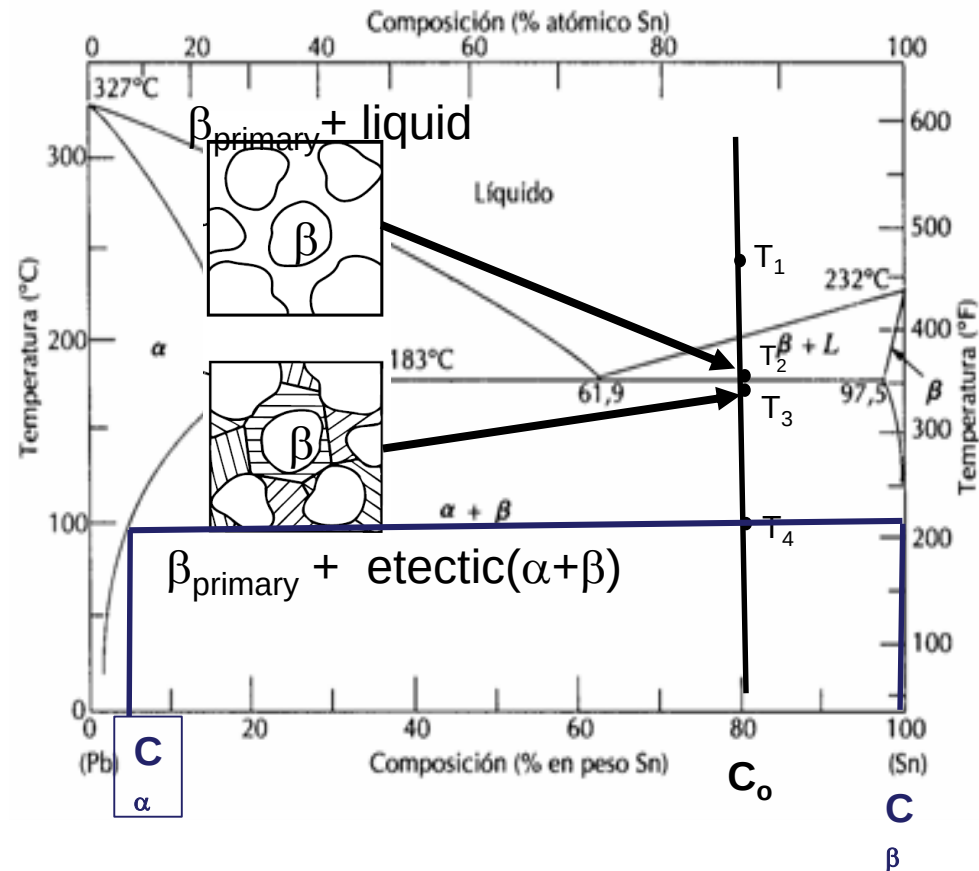
$C_{\beta \text{primary}} = 97.5\% \text{ Sn}$, $C_{\text{eutectic}} = 61.9\% \text{ Sn}$

$$\chi_{E(\alpha+\beta)} = \chi_{L@T_2} = \frac{C_\beta - C_0}{C_\beta - C_e} = \frac{97.5 - 80}{97.5 - 61.9} = 0.49$$

$$\chi_{\beta \text{primary}} = \chi_{\beta @ T_2} = 0.51$$

$$\chi_{\beta E} = \chi_{\beta \text{Total}} - \chi_{\beta \text{primary}} = 0.78 - 0.51 = 0.27$$

$$\chi_{E(\alpha+\beta)} = \chi_{\beta E} + \chi_\alpha = 0.27 + 0.22 = 0.49$$



4- Using the Pb-Sn phase diagram for an alloy composition of Pb-30%Sn, indicate at the temperatures

of $T_1 = 250^\circ\text{C}$, $T_2 = 183 + \Delta T$ °C, $T_3 = 183 - \Delta T$ °C, $T_4 = 100^\circ\text{C}$

a) The phases present, chemical composition of the phases and mass fraction.

b) The micro constituents, chemical composition and mass fractions .

•Note the agreement that exists between the data obtained in (a) and (b).

a) T_1 ; phases α and L; $C_L=36\%\text{Sn}$; $C_\alpha=12\%\text{Sn}$

$$\chi_L = \frac{C_o - C_\beta}{C_E - C_\alpha} = \frac{30 - 12}{36 - 12} = 0.75 \quad \chi_\alpha = 0.25$$

T_2 ; phases α and L; $C_L=61.9\%\text{Sn}$; $C_\alpha=19.2\%\text{Sn}$

$$\chi_L = \frac{C_o - C_\alpha}{C_E - C_\alpha} = \frac{30 - 19.2}{61.9 - 19.2} = 0.25$$

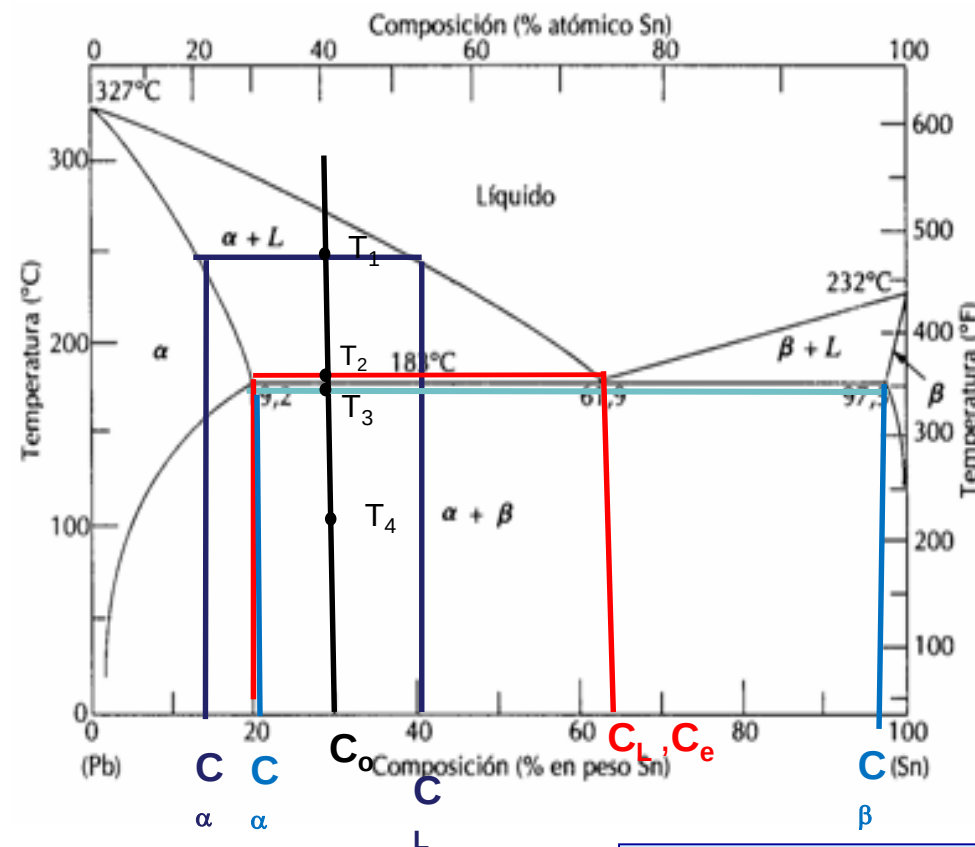
$$\chi_\alpha = 0.75 \quad (\text{Note: } \alpha \text{ phase is } \alpha_{\text{primary}})$$

T_3 ; phases α and β ; $C_\alpha = 19.2\%$; $C_\beta = 97.5\%$

$$\chi_{\alpha_{\text{Total}}} = \frac{C_\beta - C_o}{C_\beta - C_\alpha} = \frac{97.5 - 30}{97.5 - 19.2} = 0.86 \quad \chi_\beta = 0.14$$

$$\chi_{\alpha_{\text{Total}}} = \chi_{\alpha_{\text{primary}}} + \chi_{\alpha_{\text{eutectic}}}$$

$$\chi_{\alpha_{\text{primary}}} = 0.75 \quad \chi_{\alpha_{\text{eutectic}}} = 0.11$$



T4; phases α and β ; $C_\alpha = 5\% \text{Sn}$ $C_\beta = 100\% \text{Sn}$

$$\chi_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{100 - 80}{100 - 5} = 0.21 \quad \chi_{\beta \text{TOTAL}} = 0.79$$

b)

T1, T2 = microconst. are the same as phases

T3 microconst.: $\alpha_{\text{primary}} + \text{eutectic}(\alpha + \beta)$

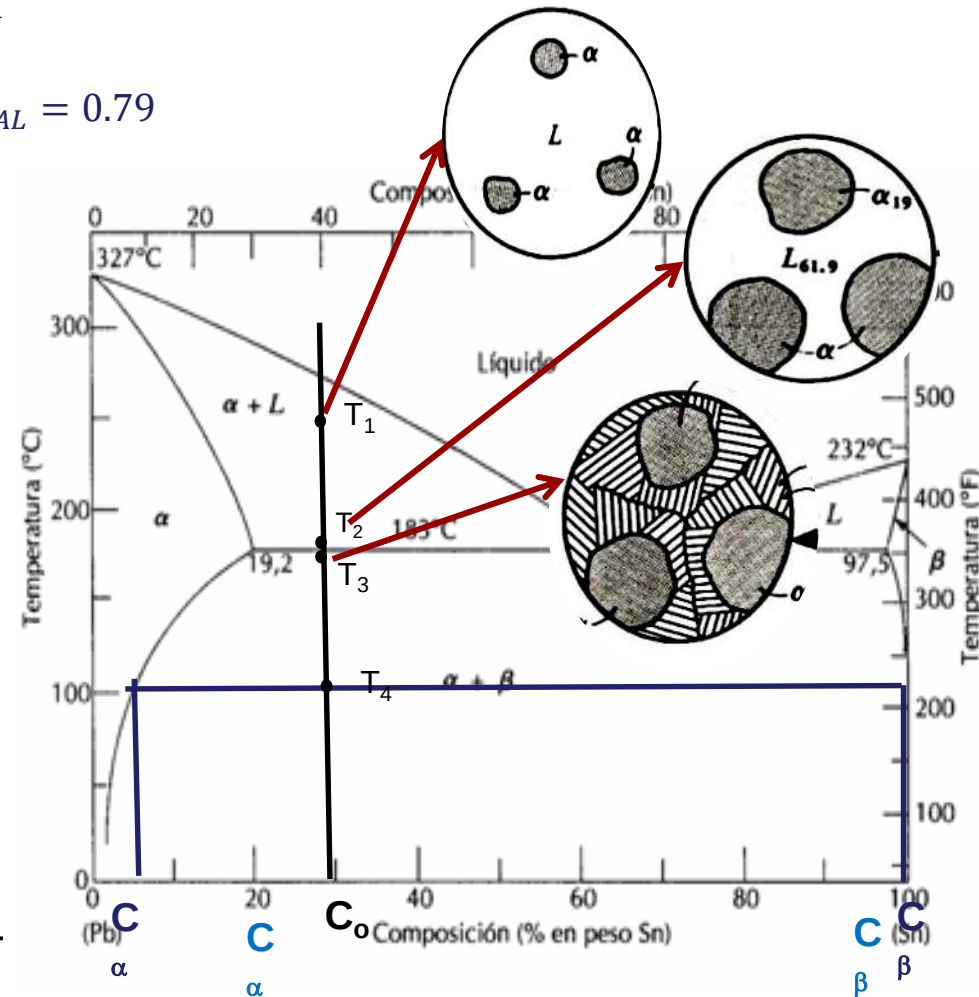
$C_{\alpha \text{primary}} = 19.2\% \text{Sn}$, $C_{\text{eutectic}} = 61.9\% \text{Sn}$

$$\chi_{\alpha \text{primary}} = \chi_{\alpha @ T_2} = \frac{C_E - C_0}{C_E - C_\alpha} = \frac{61.9 - 30}{61.9 - 19.2} = 0.75$$

$$\chi_{E(\alpha + \beta)} = \chi_{L @ T_2} = \frac{C_0 - C_\alpha}{C_E - C_\alpha} = \frac{30 - 19.2}{61.9 - 19.2} = 0.25$$

$$\chi_{\alpha_E} = 0.86 - 0.75 = 0.11$$

$$\chi_{E(\alpha + \beta)} = 0.11 + 0.14 = 0.25 \quad \chi_\beta = 0.14$$



6. Using the phase diagram of the Cu-Ag system for an alloy with composition 80wt% Cu, indicate at the temperatures of $T_1 = 900^\circ\text{C}$, $T_2 = 780 + \Delta T^\circ\text{C}$, $T_3 = 780 - \Delta T^\circ\text{C}$, $T_4 = 500^\circ\text{C}$

a) The phases present, chemical composition of the phases and mass fractions.

b) The micro constituents, chemical composition and mass fractions.

c) Note the agreement that exists between the data obtained in (a) and (b).

a)

T1; Phases L+(Cu); $C_L=58\%\text{Cu}$; $C_{(Cu)}=93\%\text{Cu}$;

$$\chi_L = \frac{C_\beta - C_o}{C_\beta - C_L} = \frac{93 - 80}{93 - 58} = 0.37$$

$$\chi_{(Cu)} = 1 - 0.37 = 0.63$$

T2; phase L+(Cu); $C_L=28.1\%\text{Cu}$; $C_{(Cu)}=92\%\text{Cu}$;

$$\chi_L = \frac{C_\beta - C_o}{C_\beta - C_L} = \frac{92 - 80}{92 - 28.1} = 0.19$$

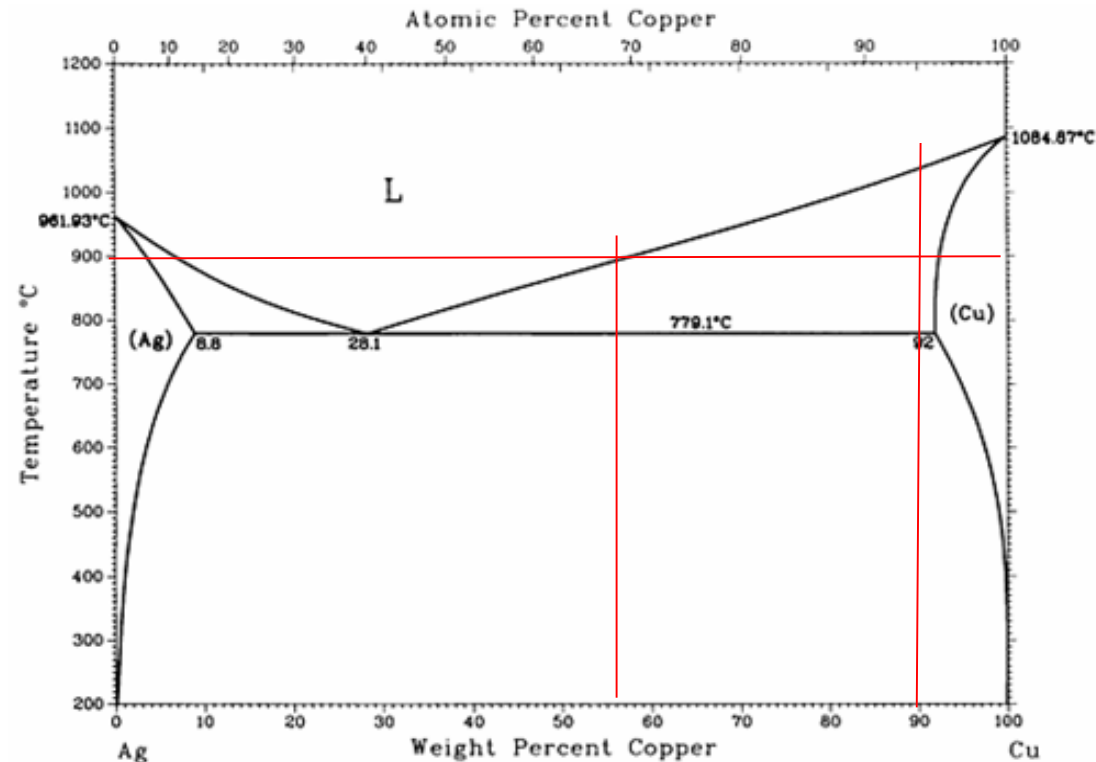
$$\chi_{(Cu)} = 1 - 0.19 = 0.81$$

T3; phases (Cu) + (Ag);

$C_{(Ag)} = 8.8\%$; $C_{(Cu)} = 92\%$

$$\chi_{(Ag)} = \frac{C_\beta - C_o}{C_\beta - C_\alpha} = \frac{92 - 80}{92 - 8.8} = 0.14$$

$$\chi_{(Cu)} = 0.86$$



b) T3: (Cu)_{primary} + eutectic

$$C_{\text{Cu primary}} = 92\%, C_{\text{eutectic}} = 28.1$$

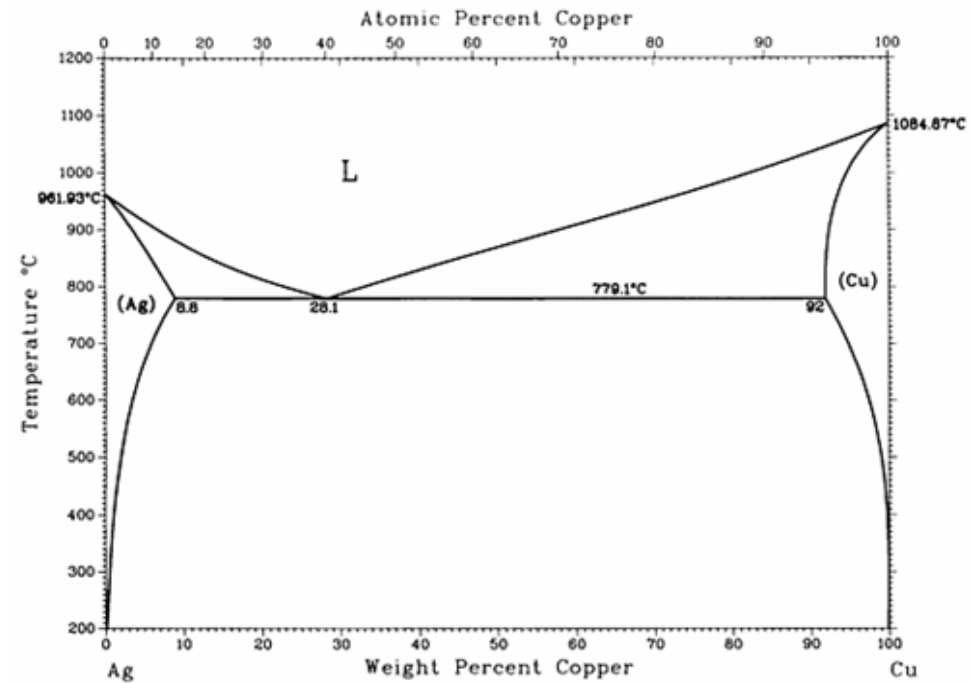
$$\begin{aligned} \chi_{\text{Cu primary}} &= \chi_{\text{Cu}@T_2} = \frac{C_0 - C_E}{C_{\text{Cu}} - C_E} \\ &= \frac{80 - 28.1}{92 - 28.1} = 0.81 \end{aligned}$$

$$\begin{aligned} \chi_E &= \chi_{\text{L}@T_2} = \frac{C_{\text{Cu}} - C_0}{C_{\text{Cu}} - C_E} = \frac{92 - 80}{92 - 28.1} \\ &= 0.19 \end{aligned}$$

$$\chi_{\text{Cu Total}} = \chi_{\text{Cu primary}} + \chi_{\text{Cu E}}$$

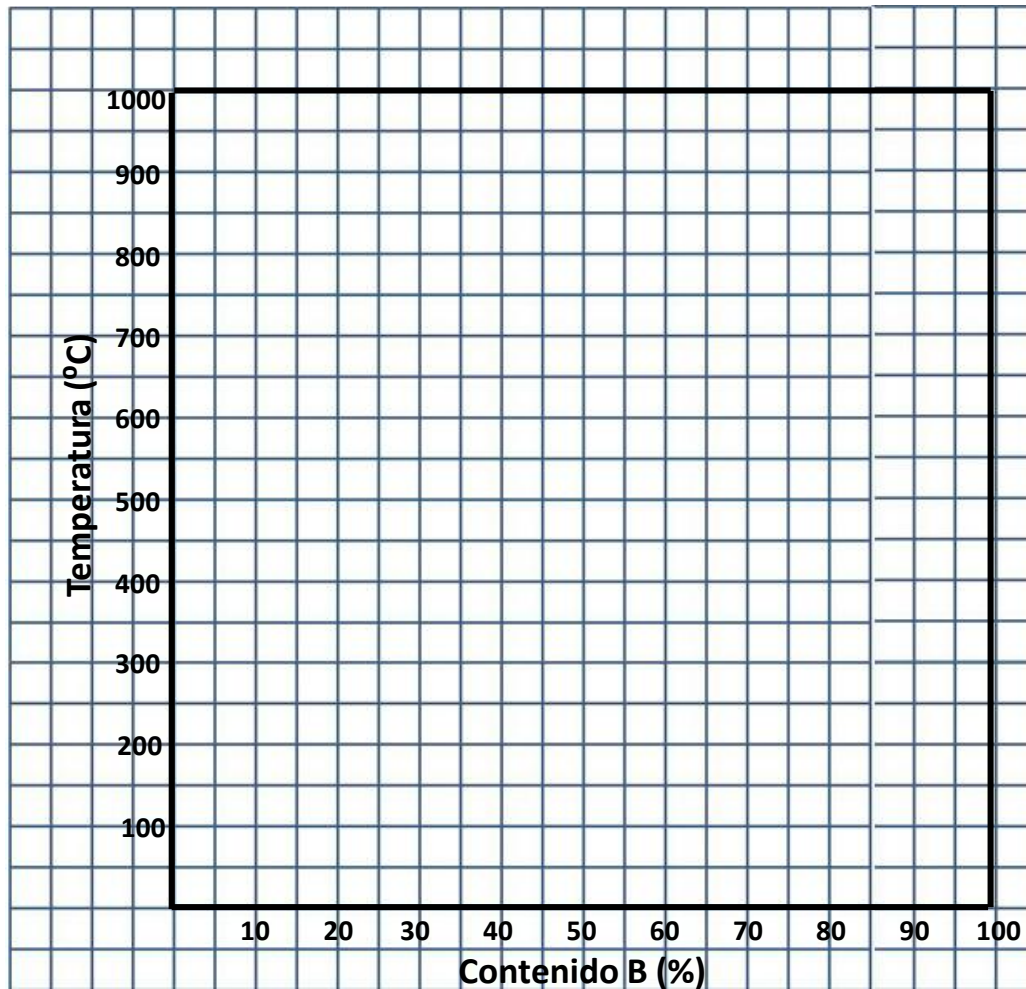
$$\chi_{\text{Cu E}} = 0.86 - 0.81 = 0.04$$

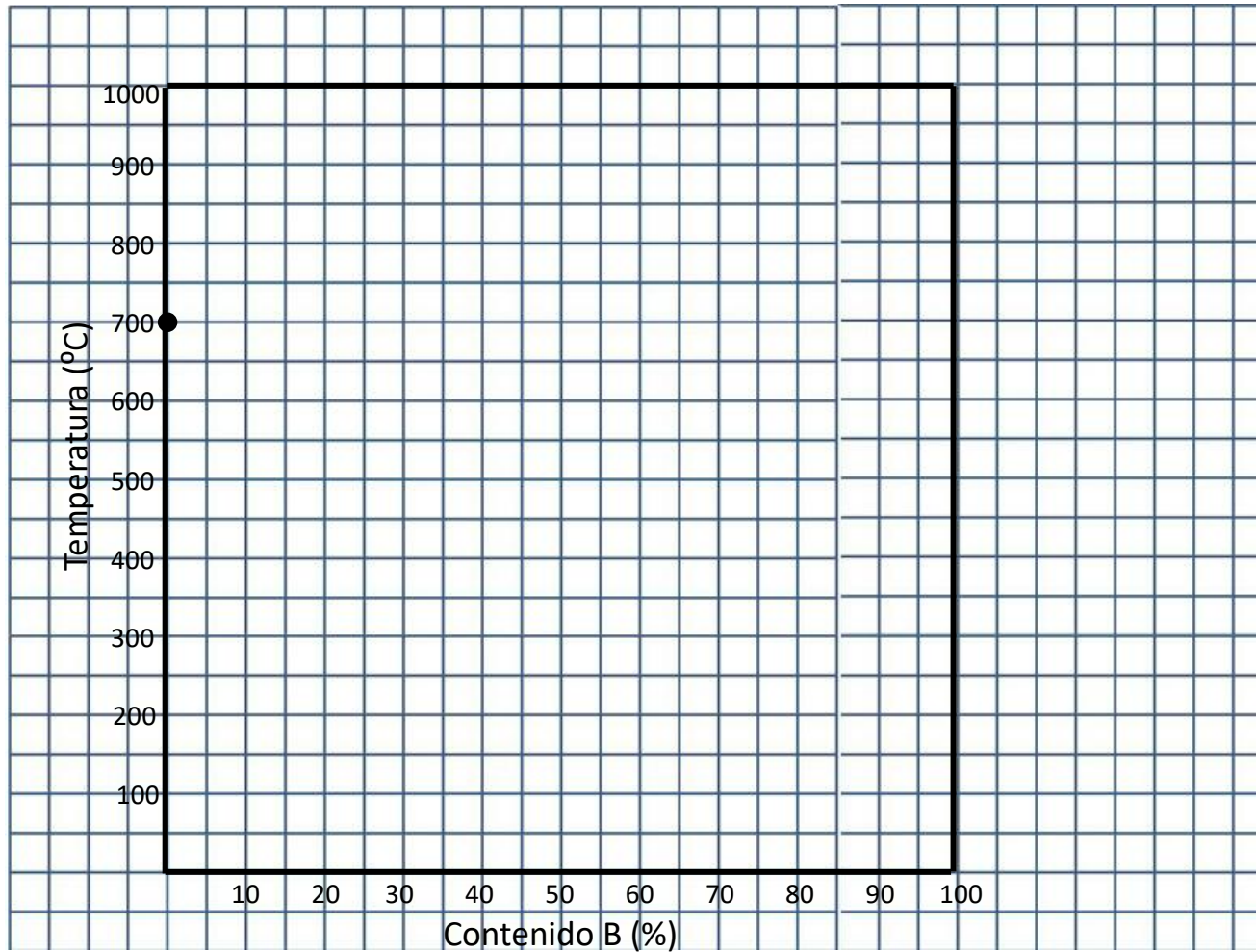
$$\chi_{\text{Ag eutectic}} = \chi_{\text{Ag Total}} = 0.14$$

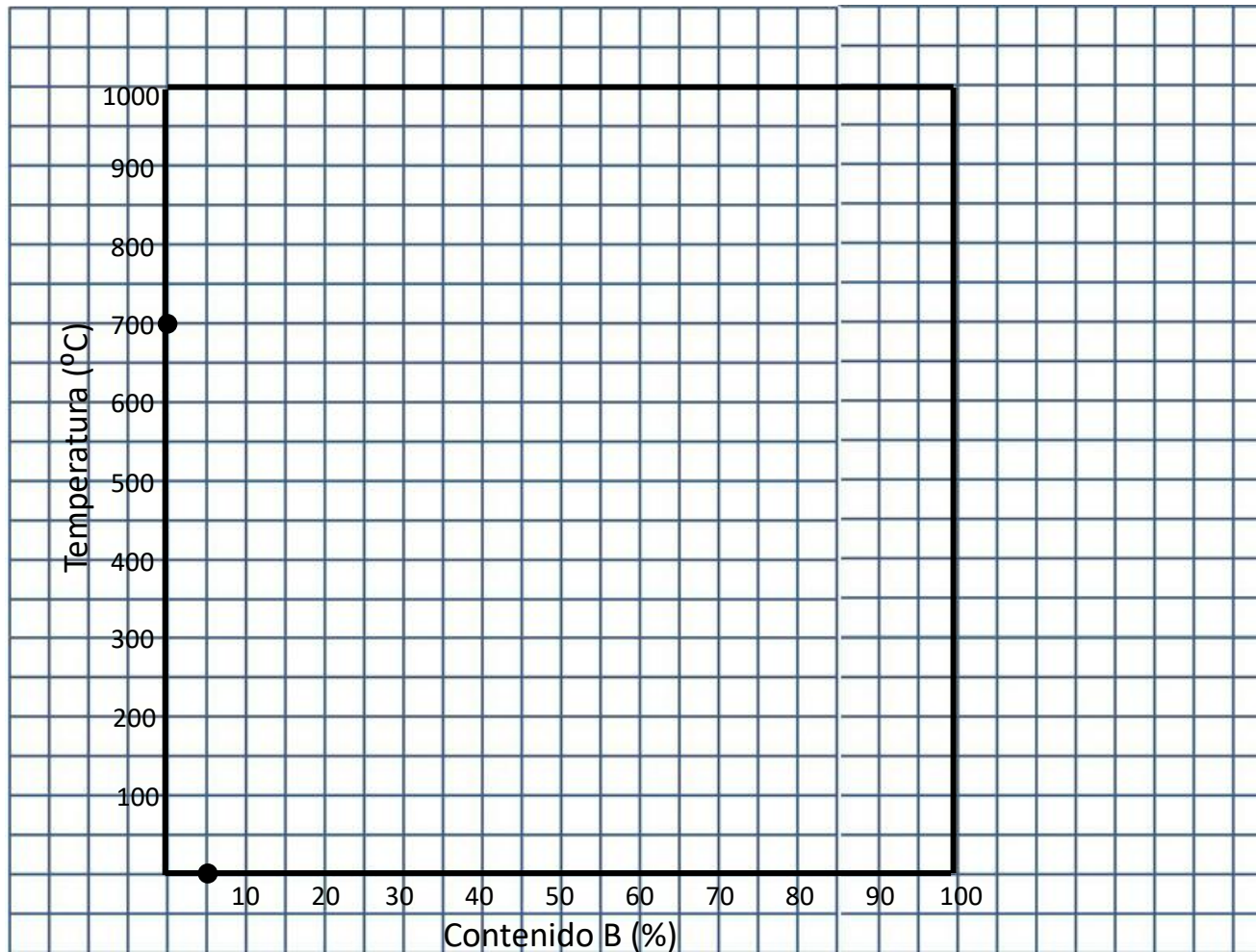


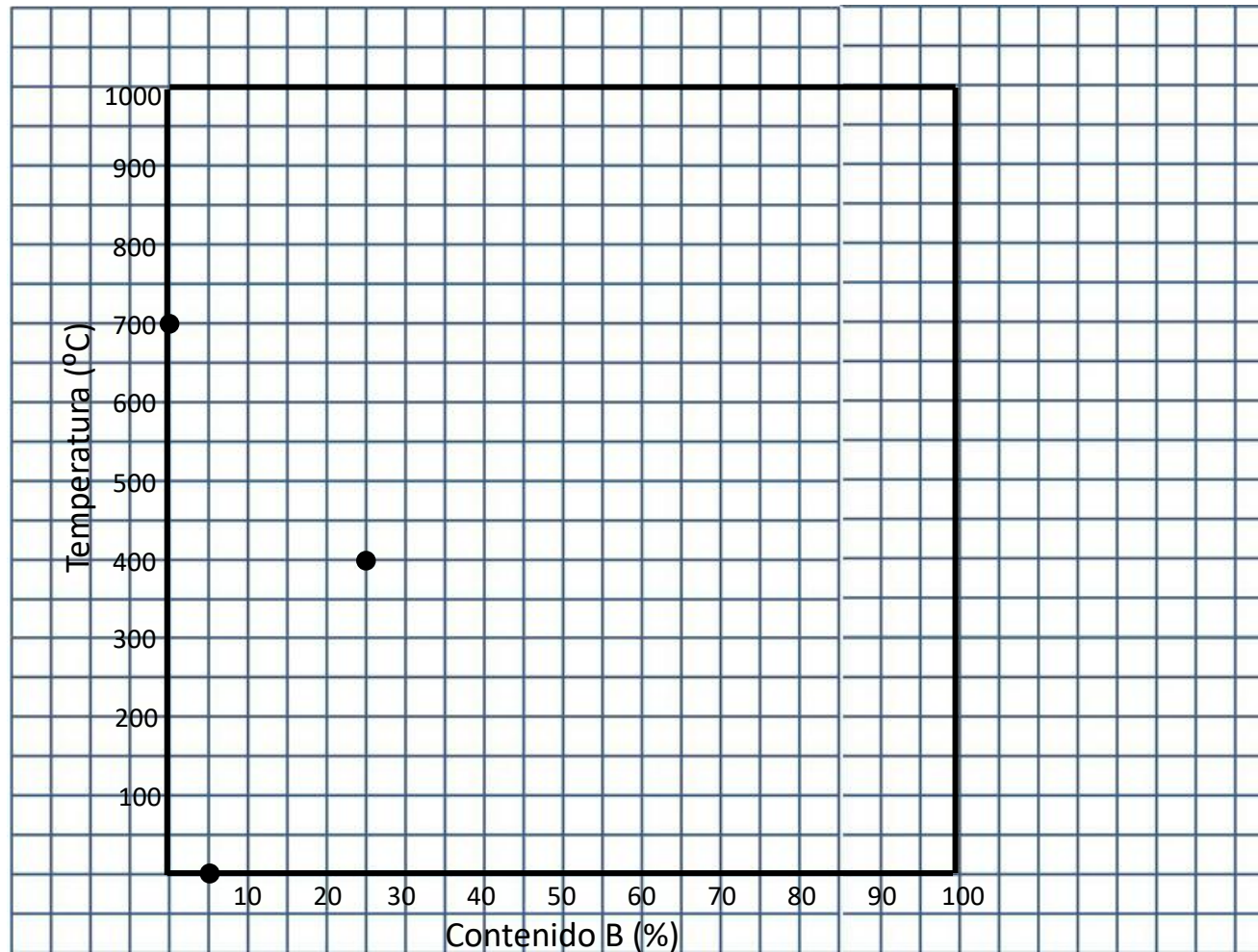
Two metals, A and B, whose melting temperatures are 700°C and 600°C , respectively, show partial solubility between them. One can dissolve 5% of B at 0°C and the maximum solubility of B in A is 25%B which takes place at 400°C . The maximum solubility of A in B is 20 %A and takes place at 400°C . B can dissolve 10 %A at 0°C . At 400°C there is an eutectic reaction with 65% B.

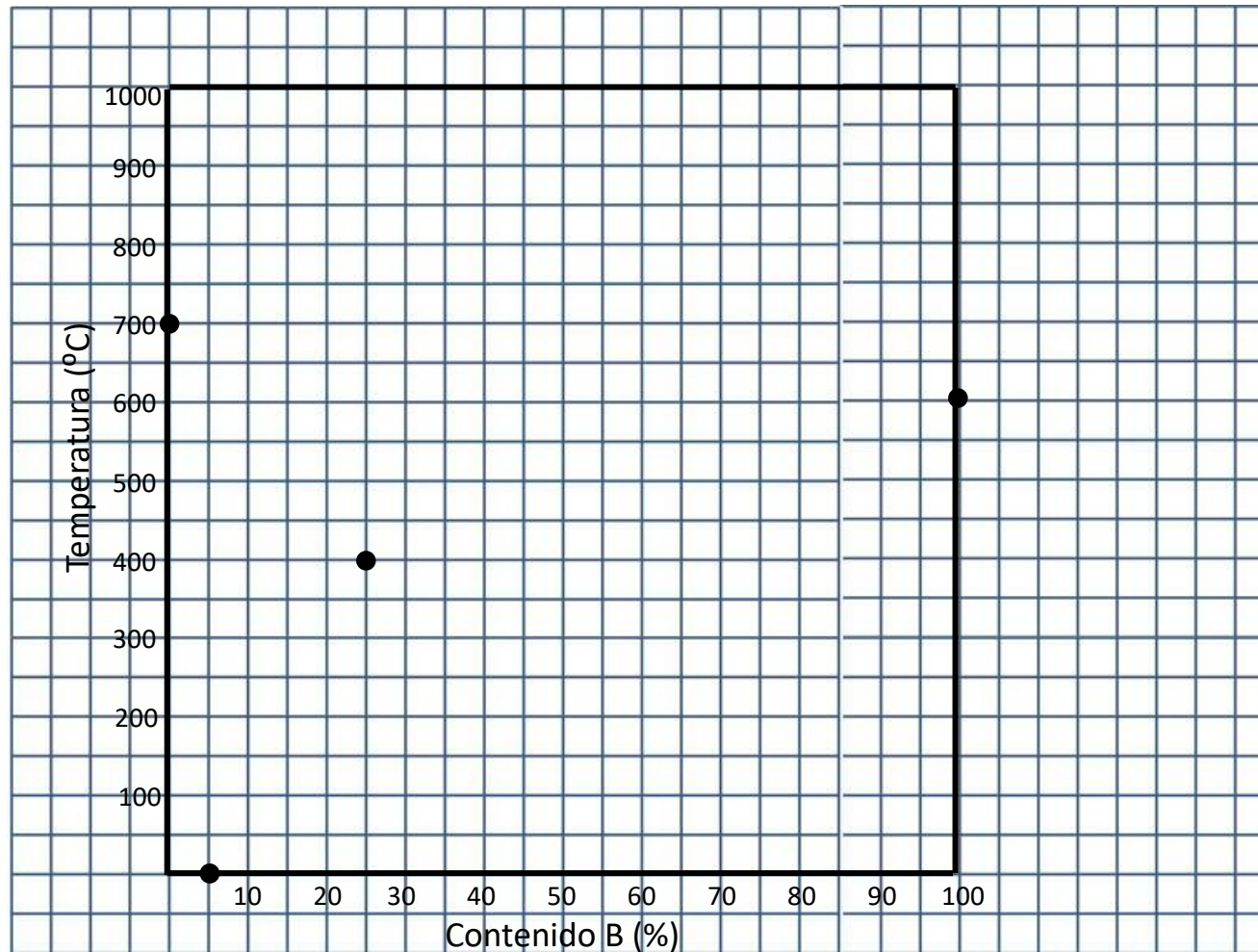
- Draw the equilibrium diagram including the phases present in each region
- For an alloy with 40%B, determine the compositions of the phases and their relative amounts at 450°C , $400^{\circ}\text{C}+\Delta T$ and $400^{\circ}\text{C}-\Delta T$.
- Draw the microstructure of the 40 %B alloy at $400^{\circ}\text{C}-\Delta T$. Indicate the microconstituents and their mass fractions. Calculate the mass fraction of the phases present in each microconstituent.

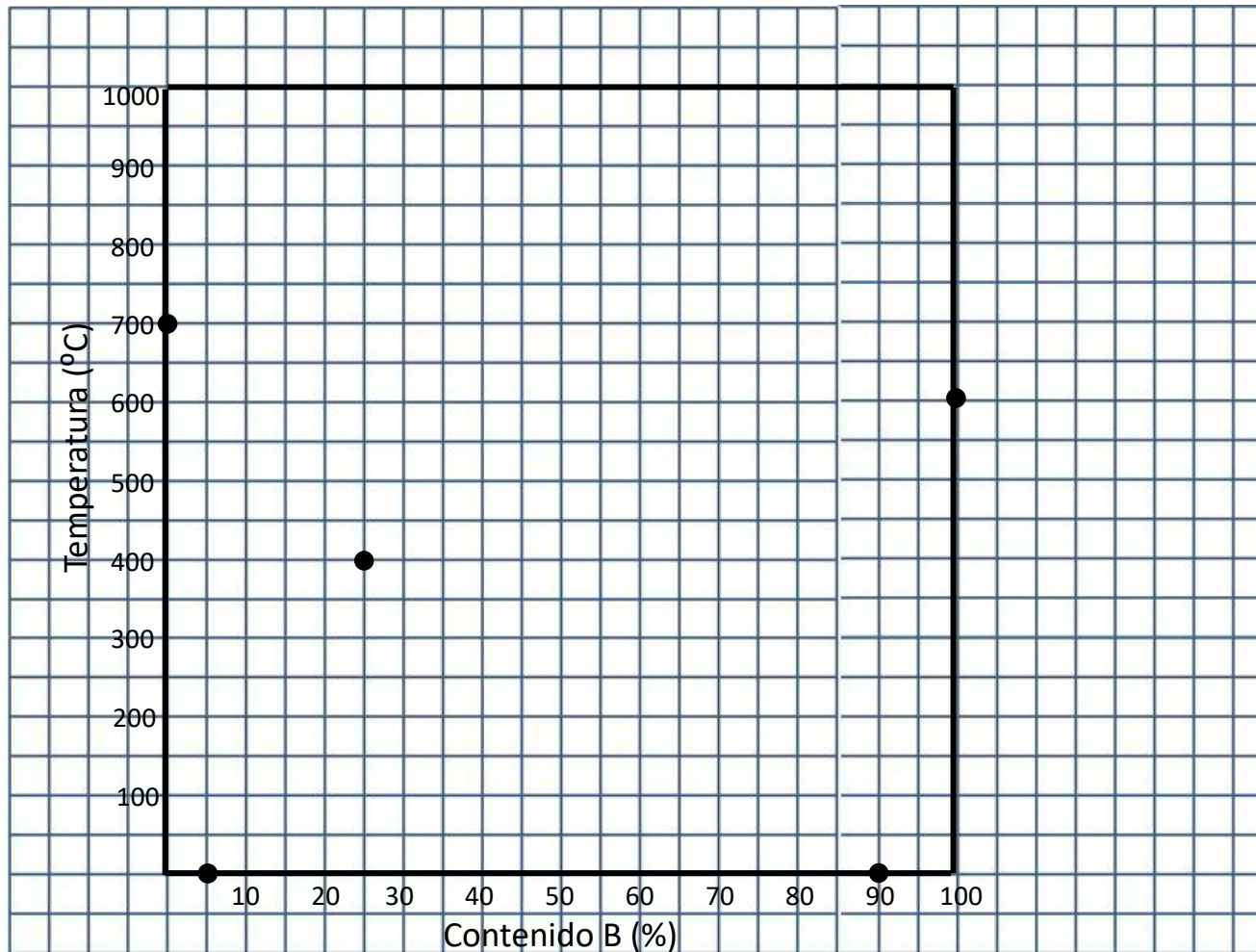


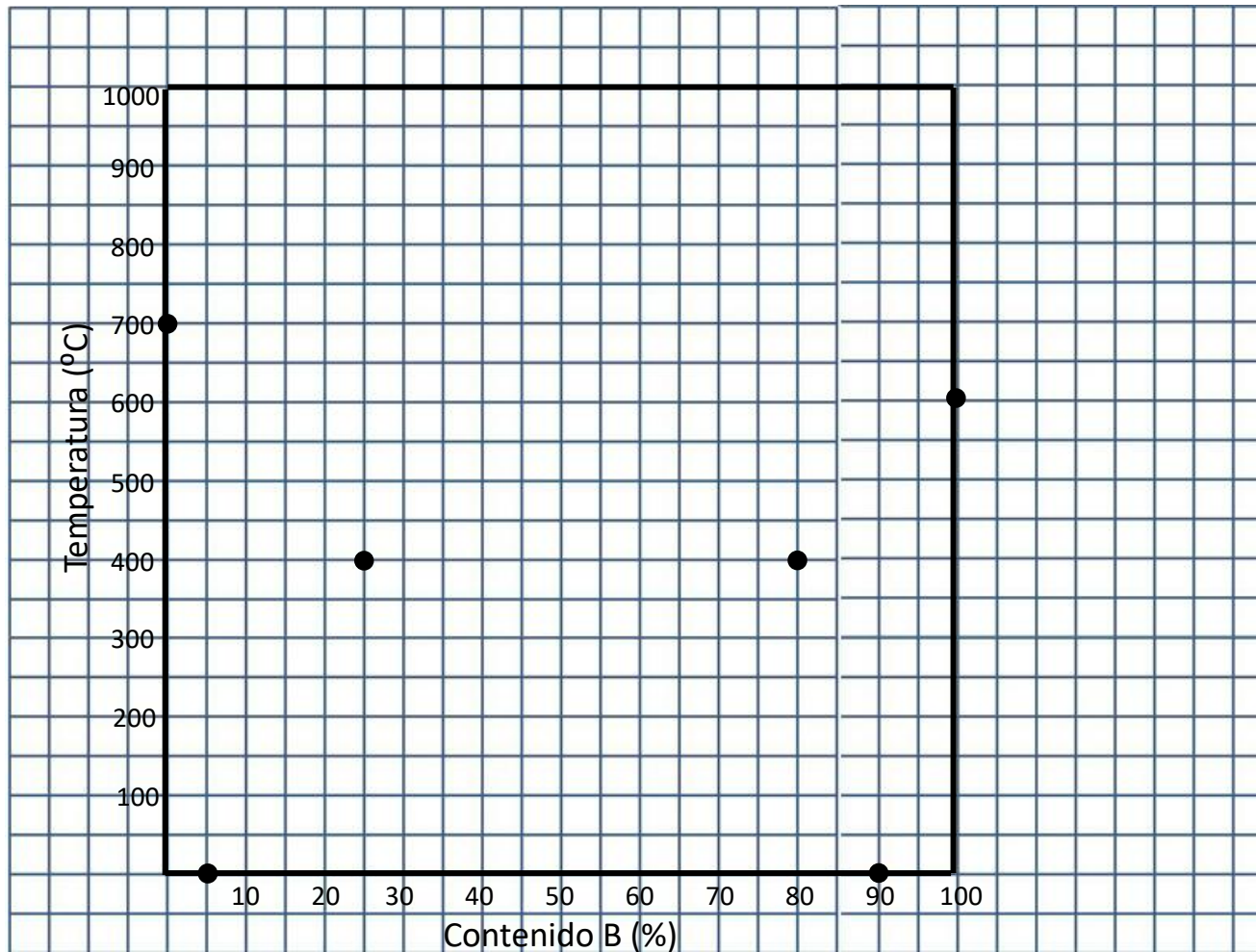


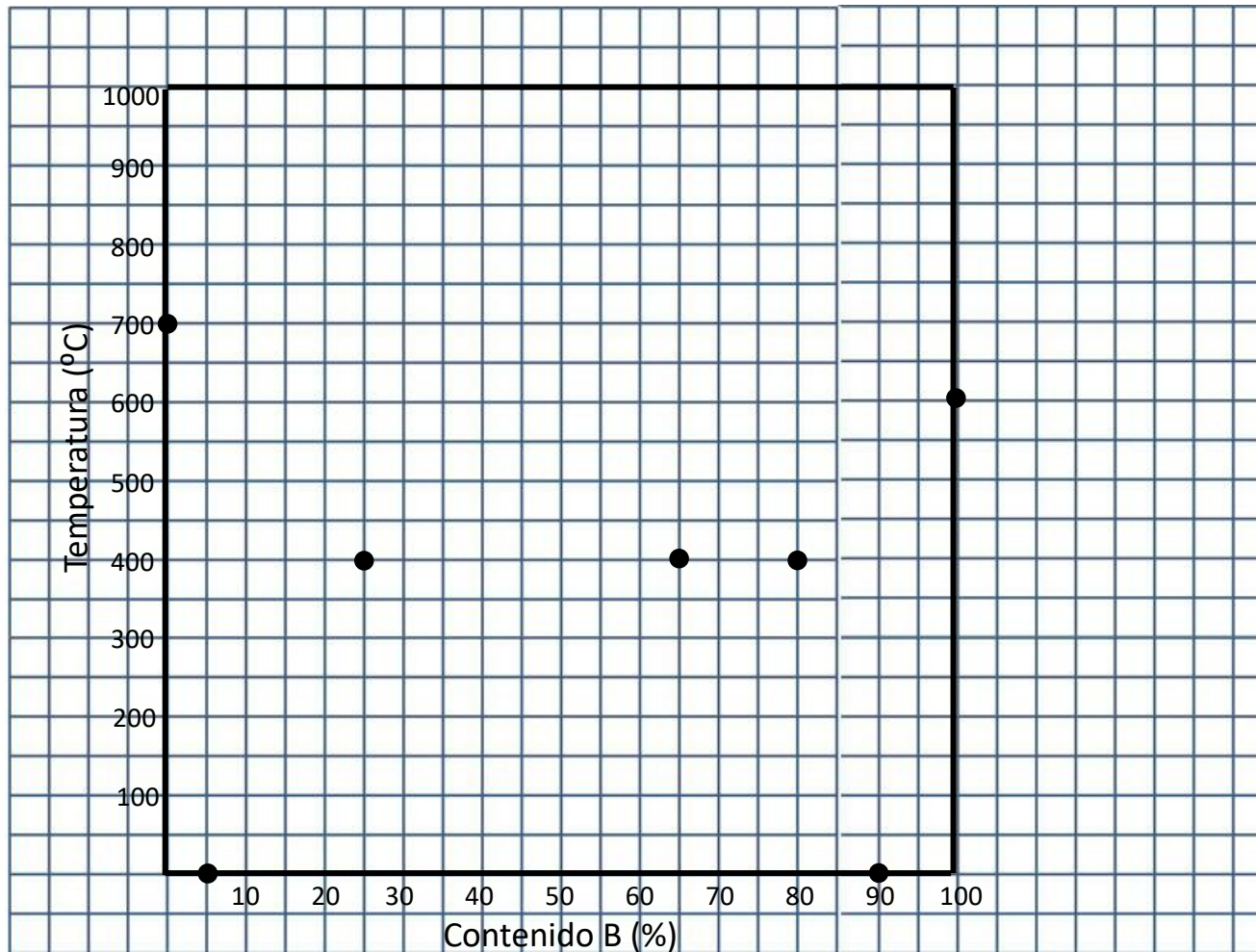


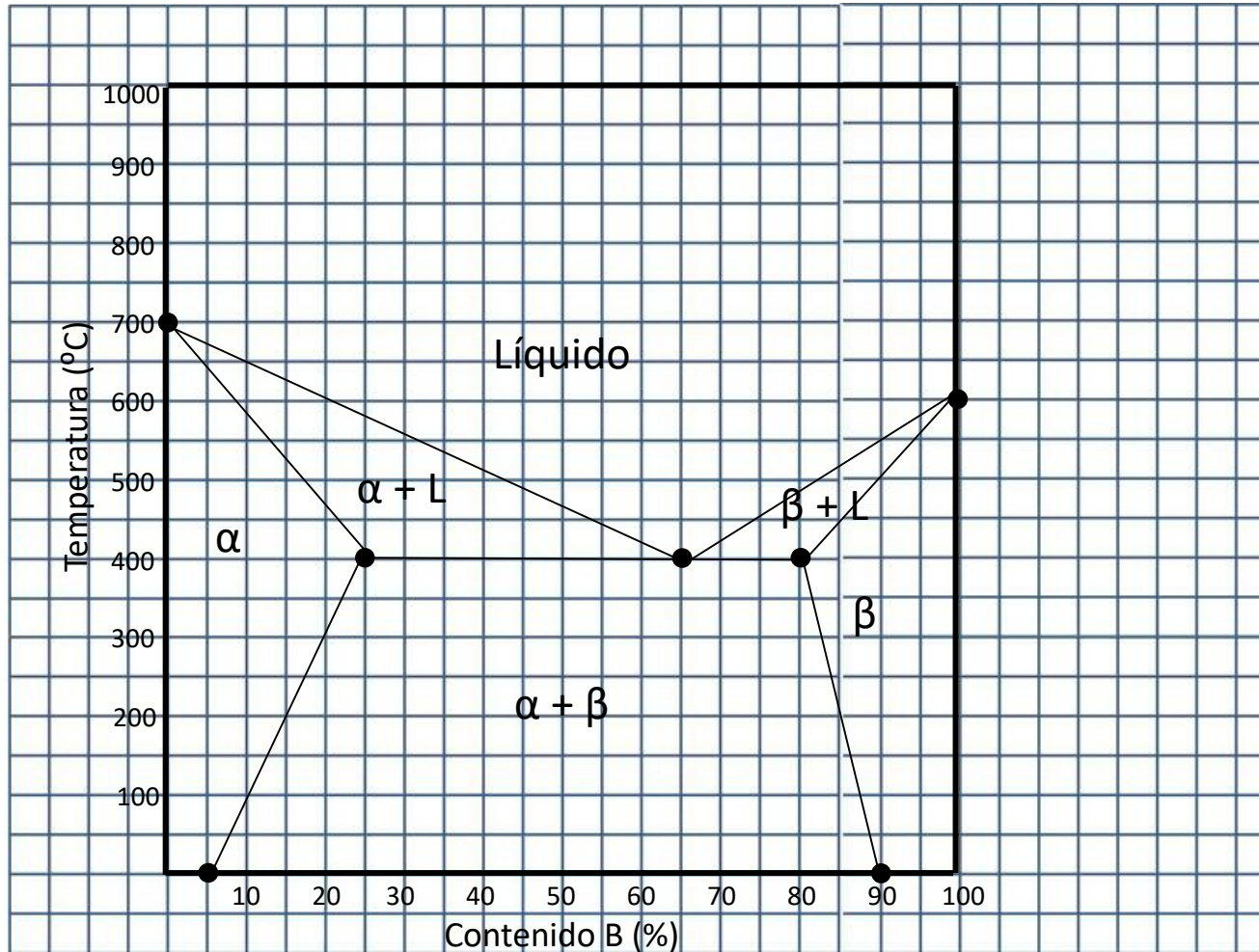


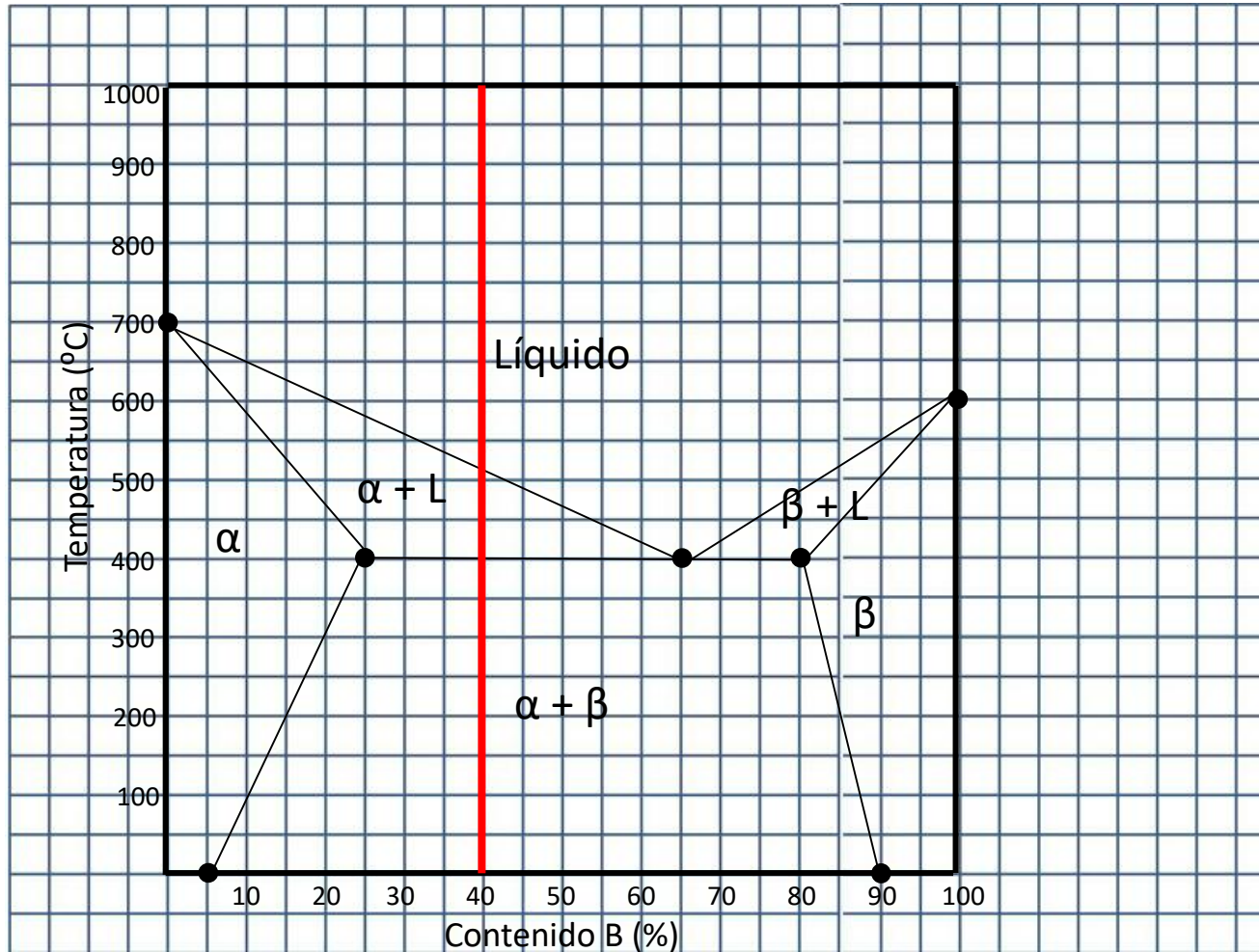


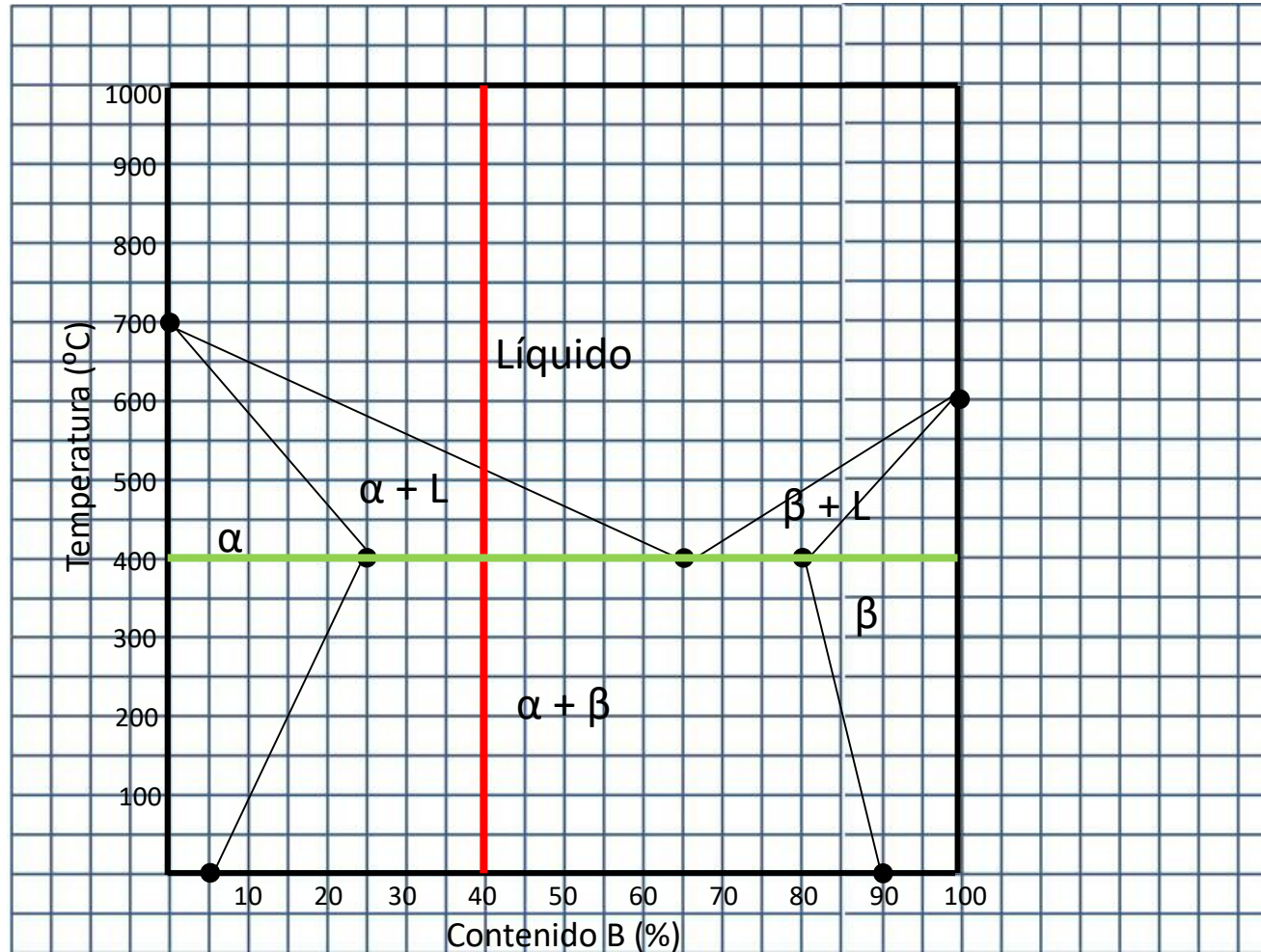


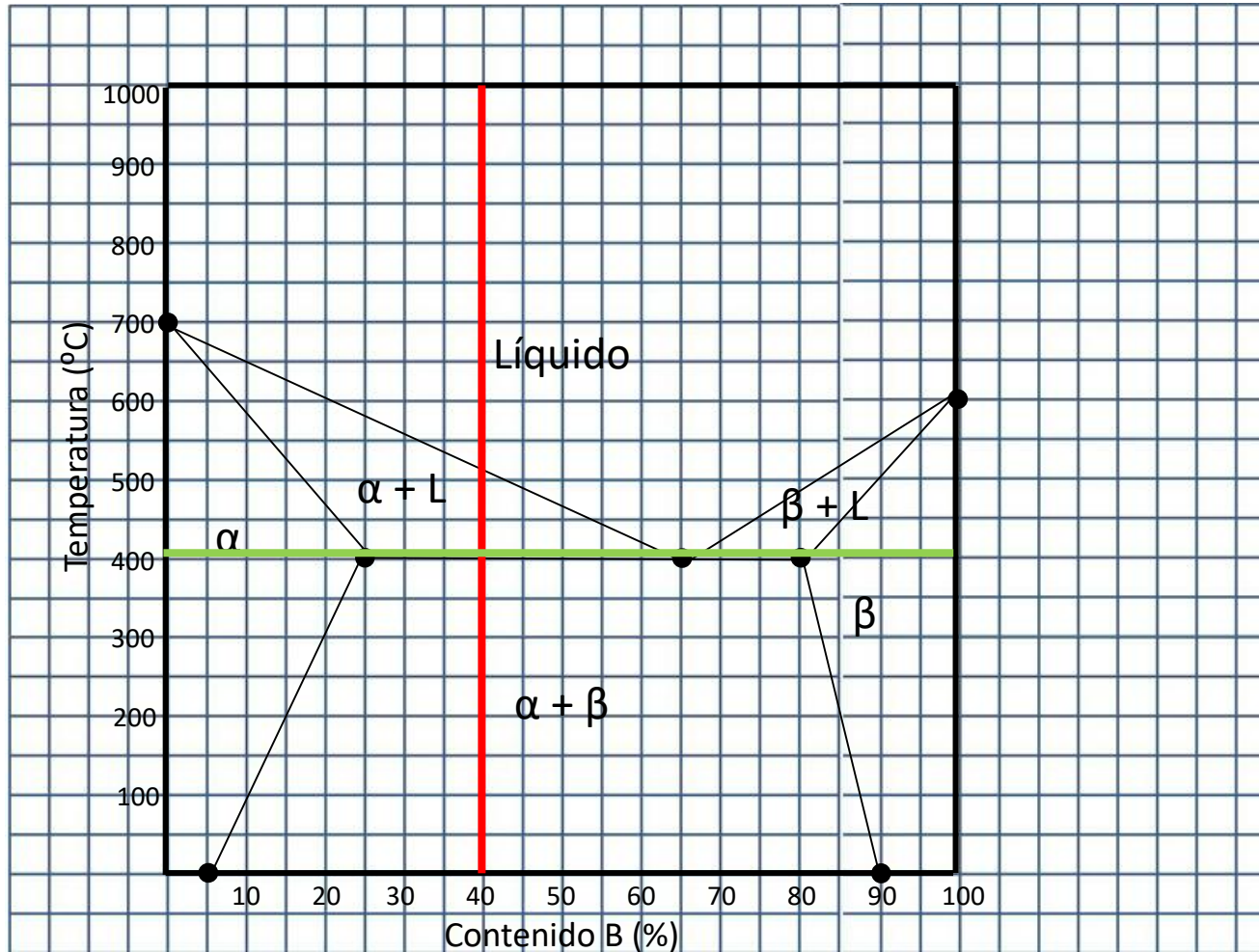








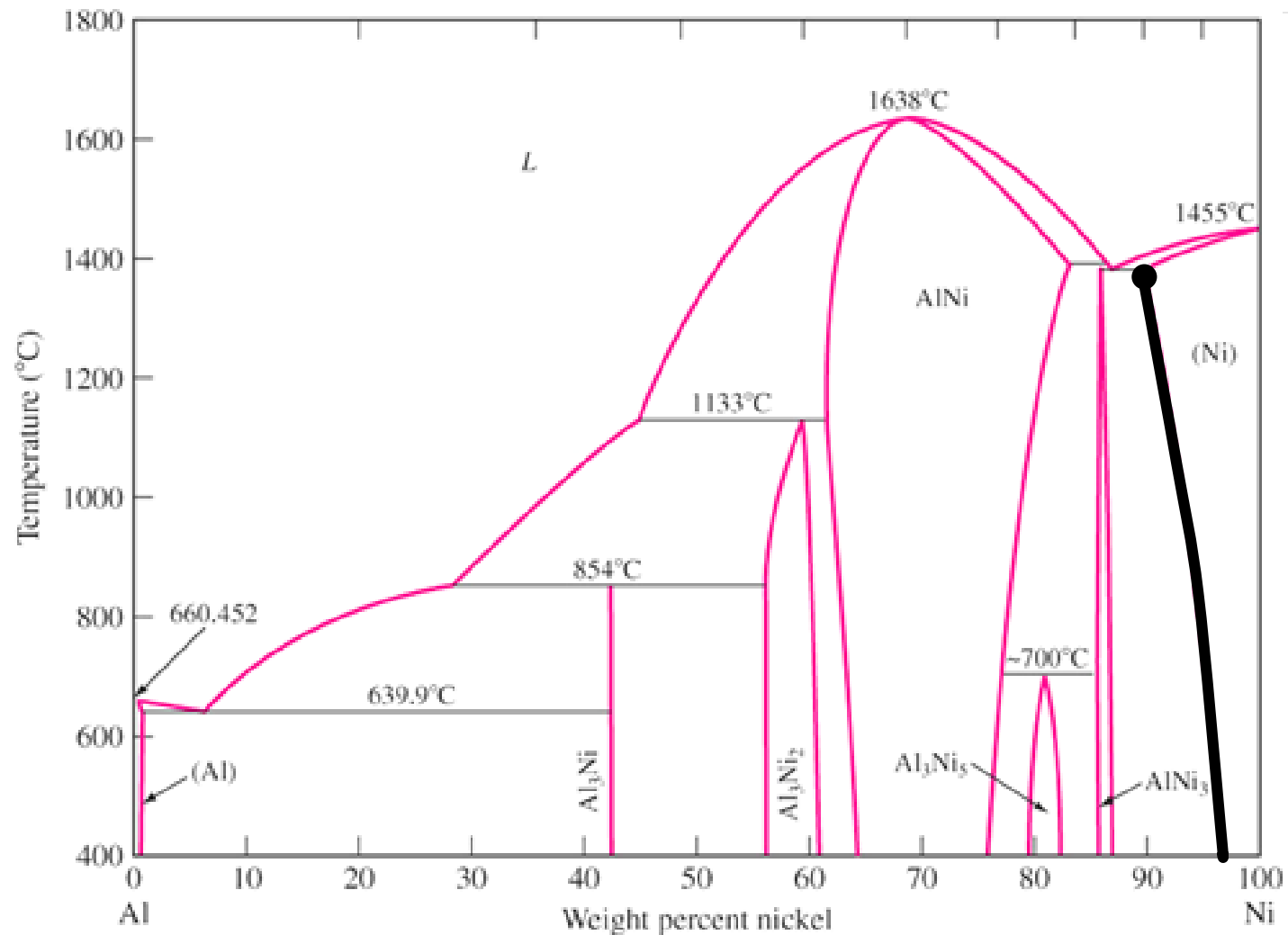


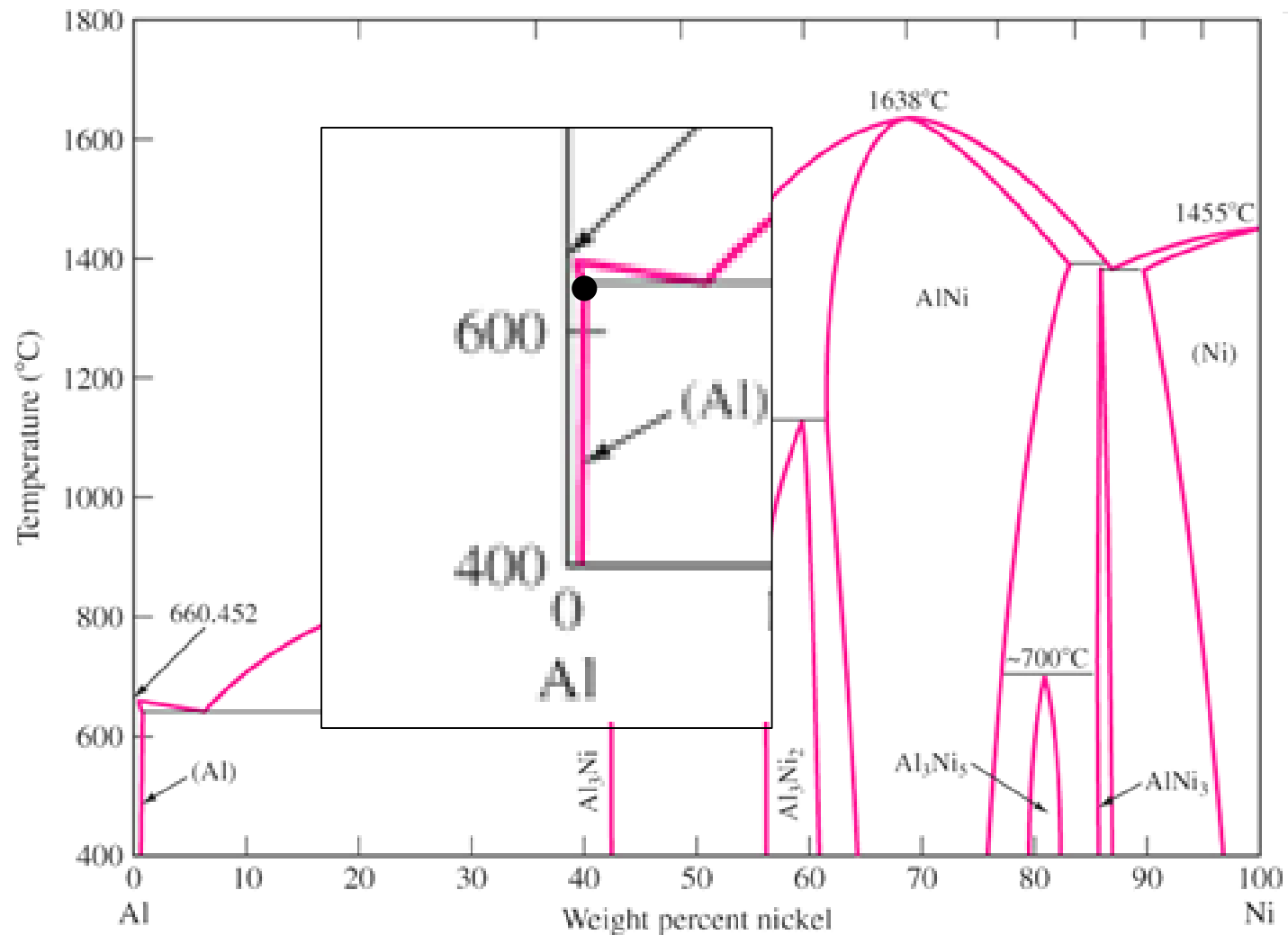


Given the Al-Ni equilibrium phase diagram, answer the next questions.

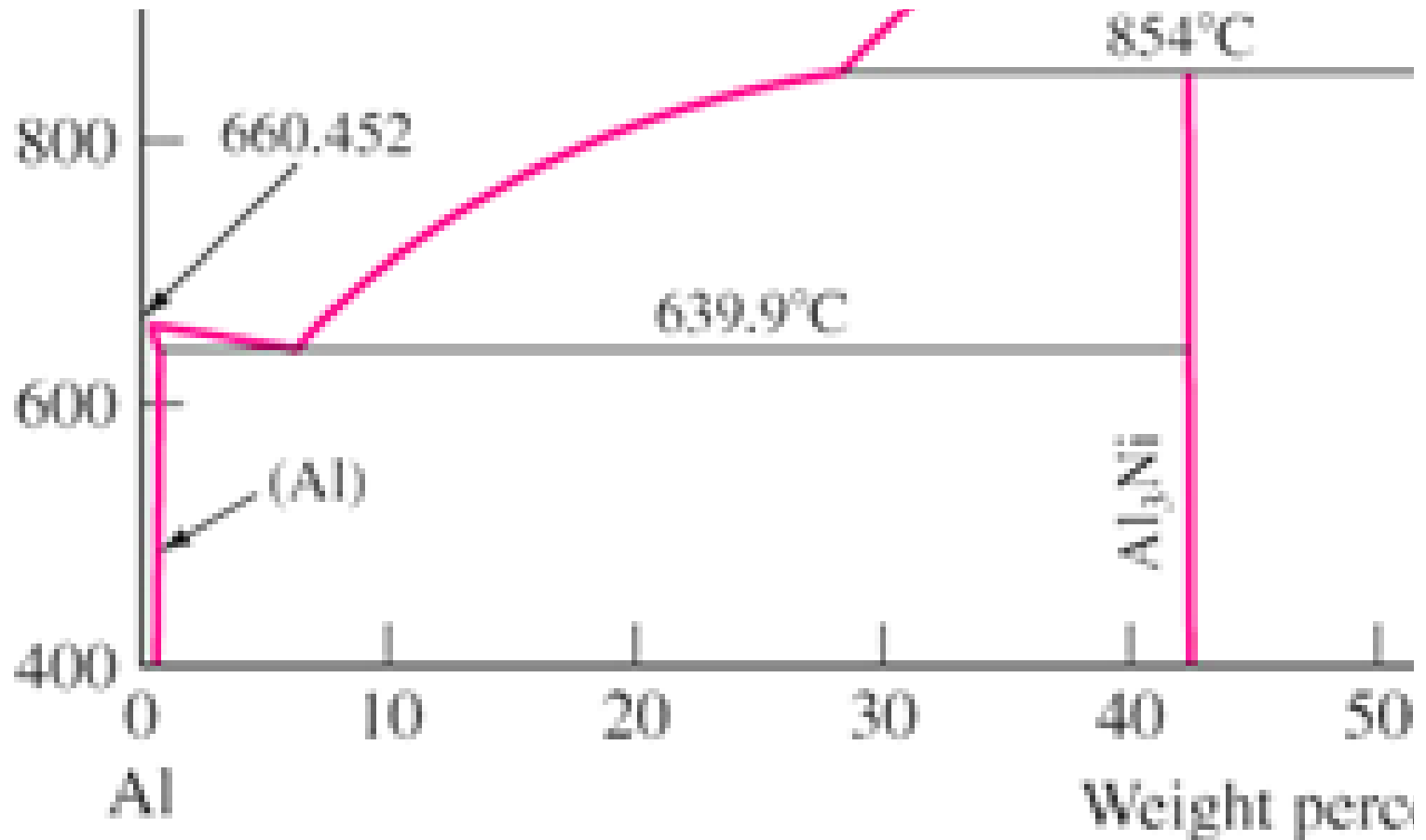
- a) What is the maximum solubility of Al in Ni in solid state? What is the maximum solubility of Ni in Al in solid state?
- b) Indicate the invariant reactions.
- c) For an alloy with 20 %Ni at 400 °C, indicate:
 - the phases present: as well as their composition and mass fraction.
 - Microconstituents present: as well as their mass fraction and the mass fraction of the phases present in each microconstituent.
 - Draw the microstructure after the cooling

PROBLEM 9

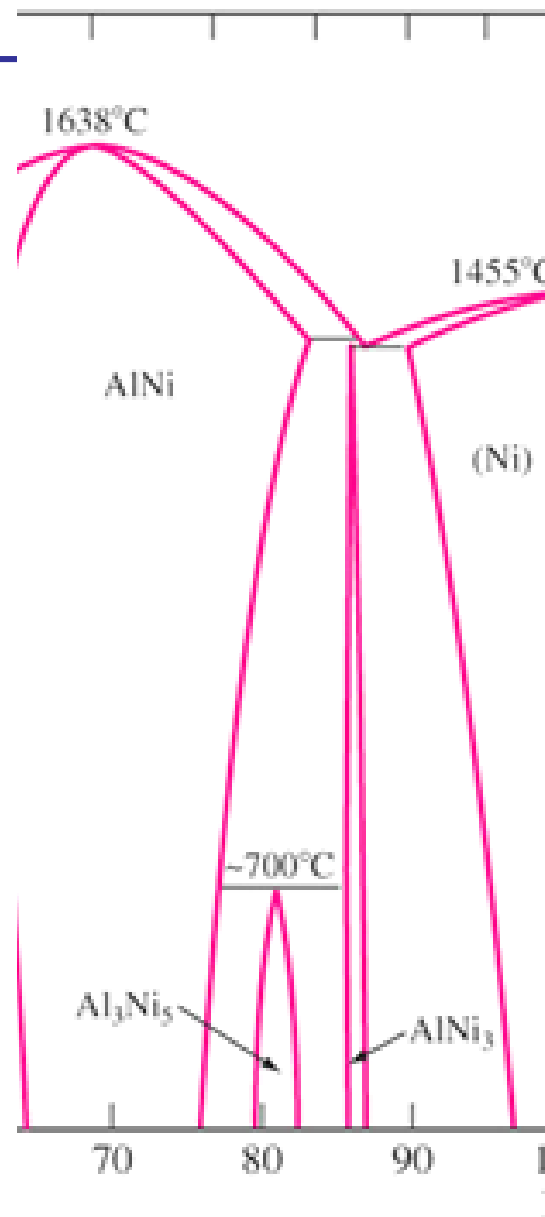




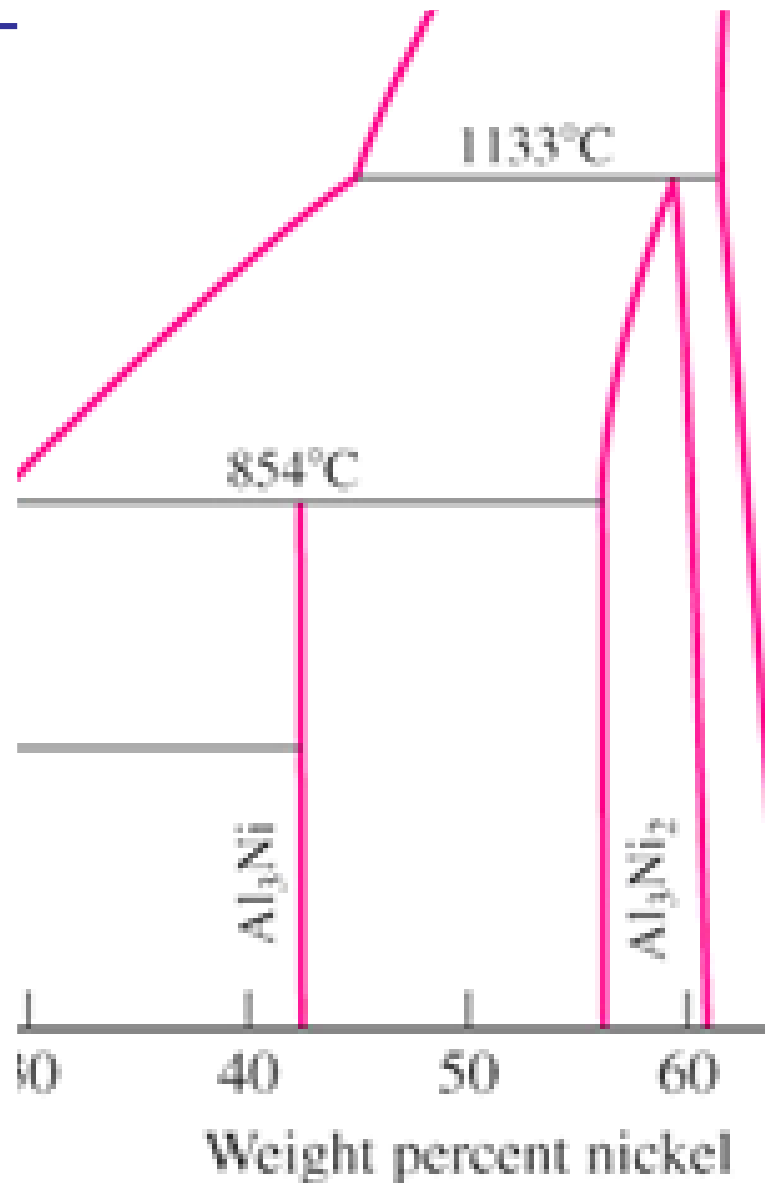
PROBLEM 9



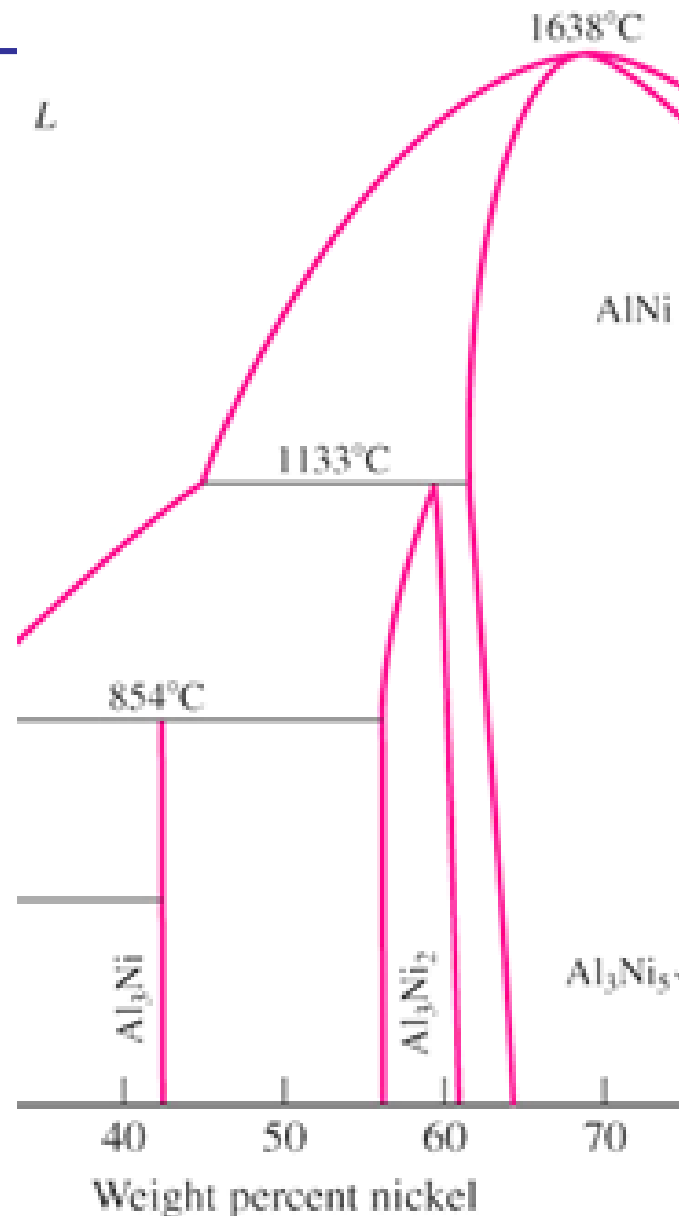
PROBLEM 9

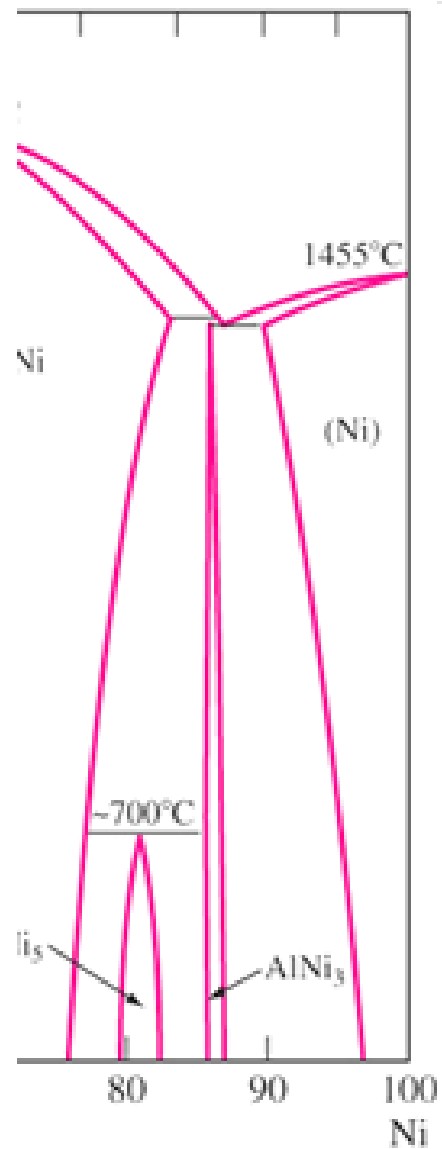


PROBLEM 9

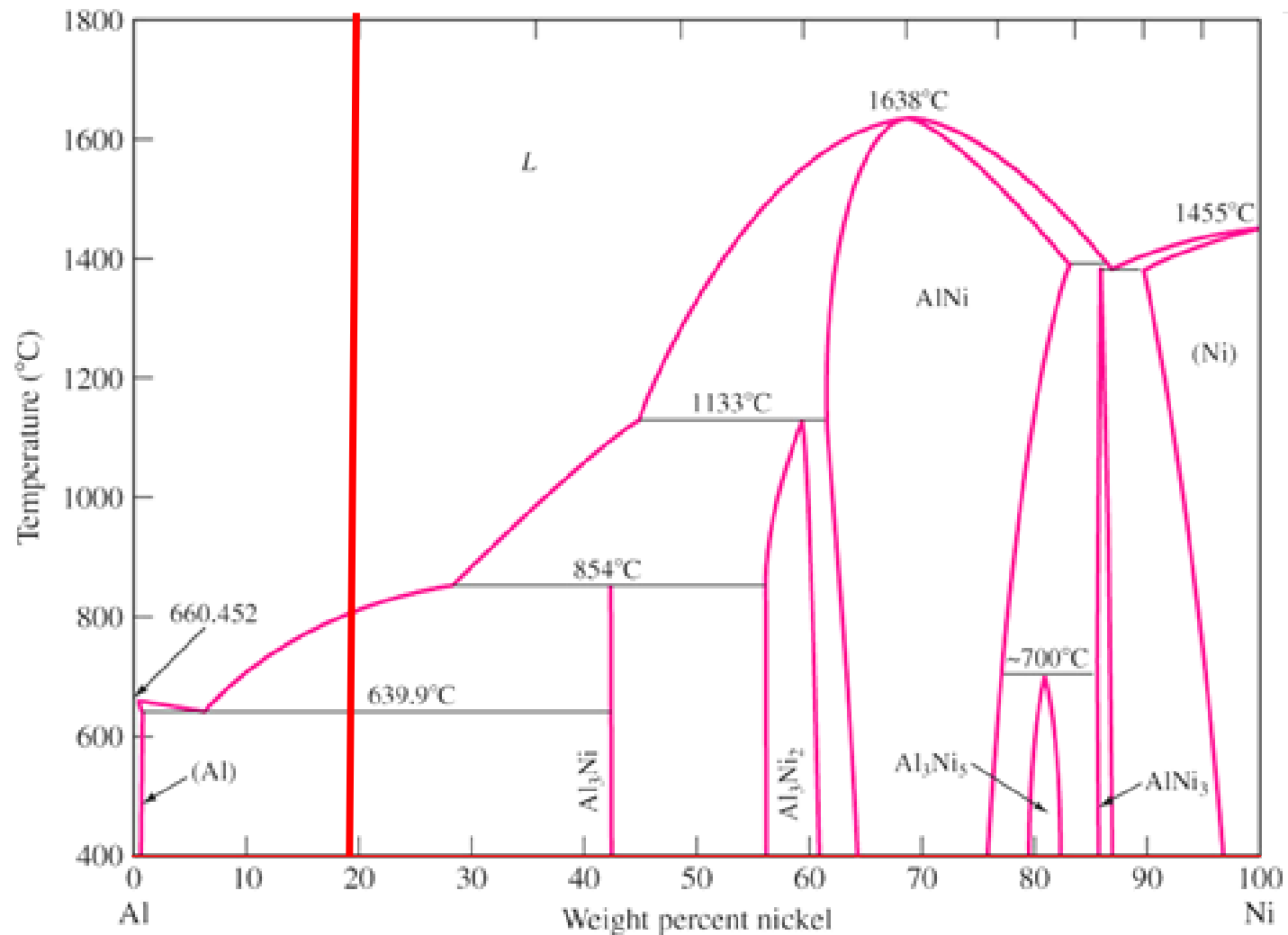


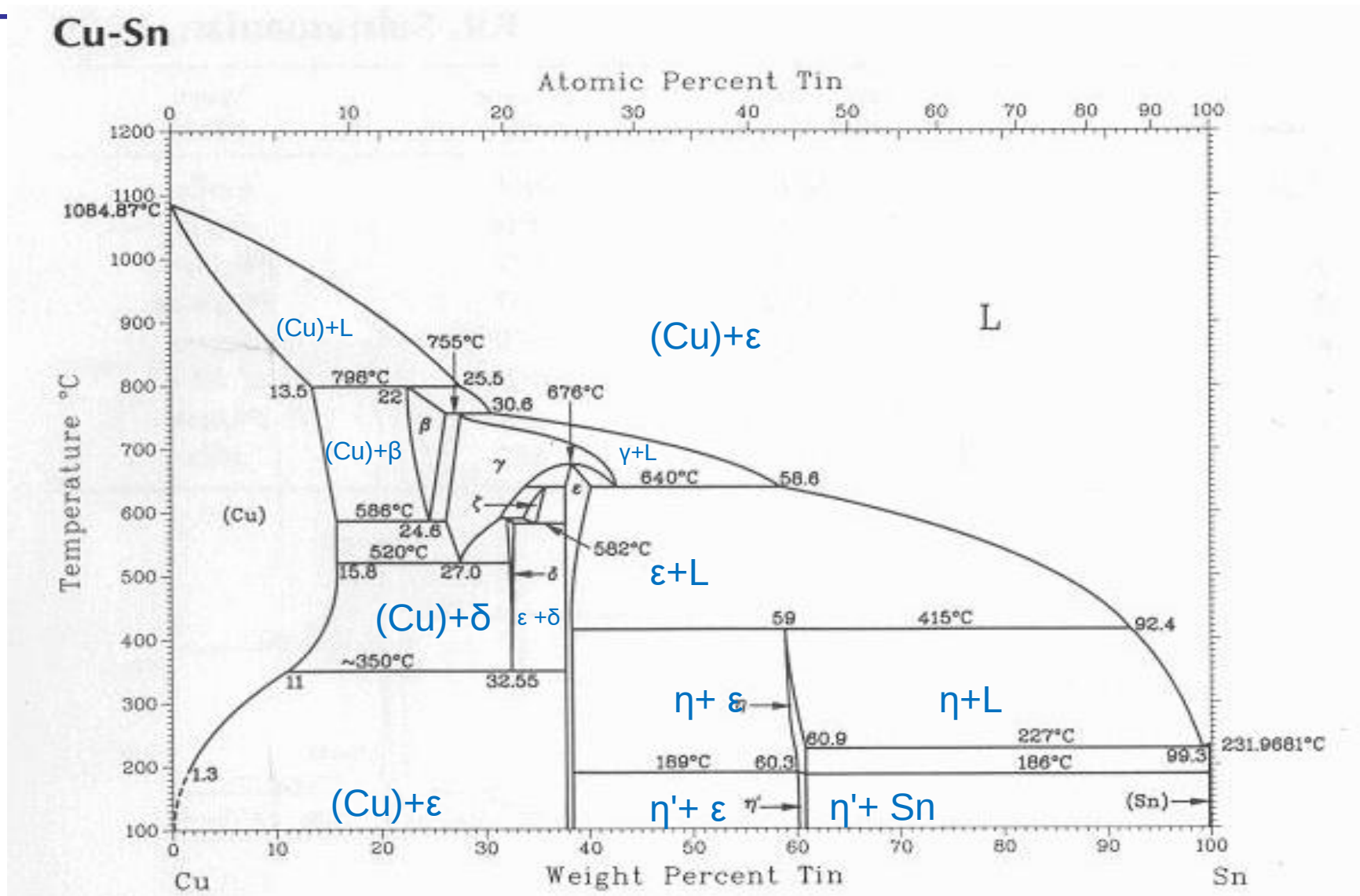
PROBLEM 9

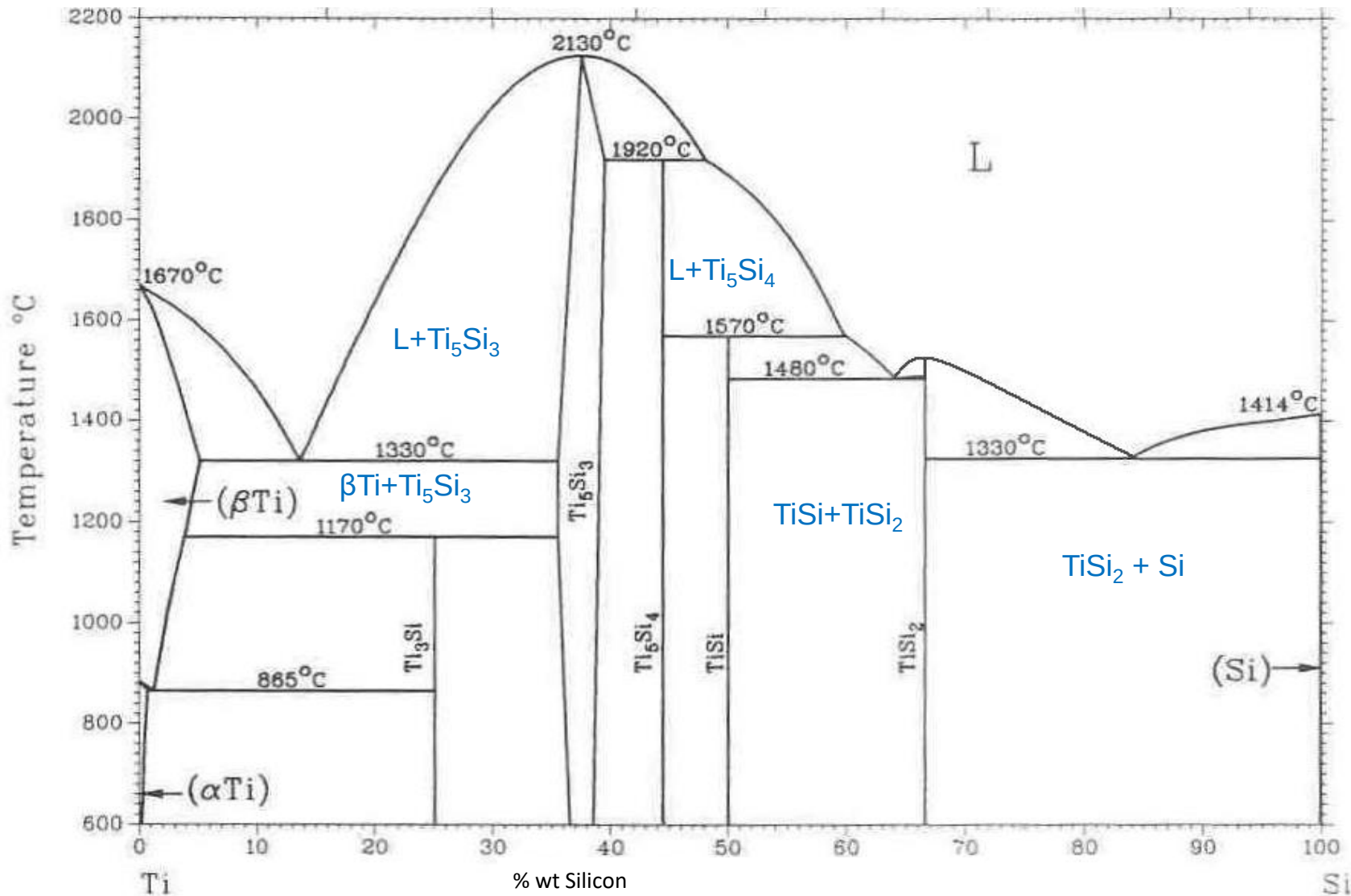


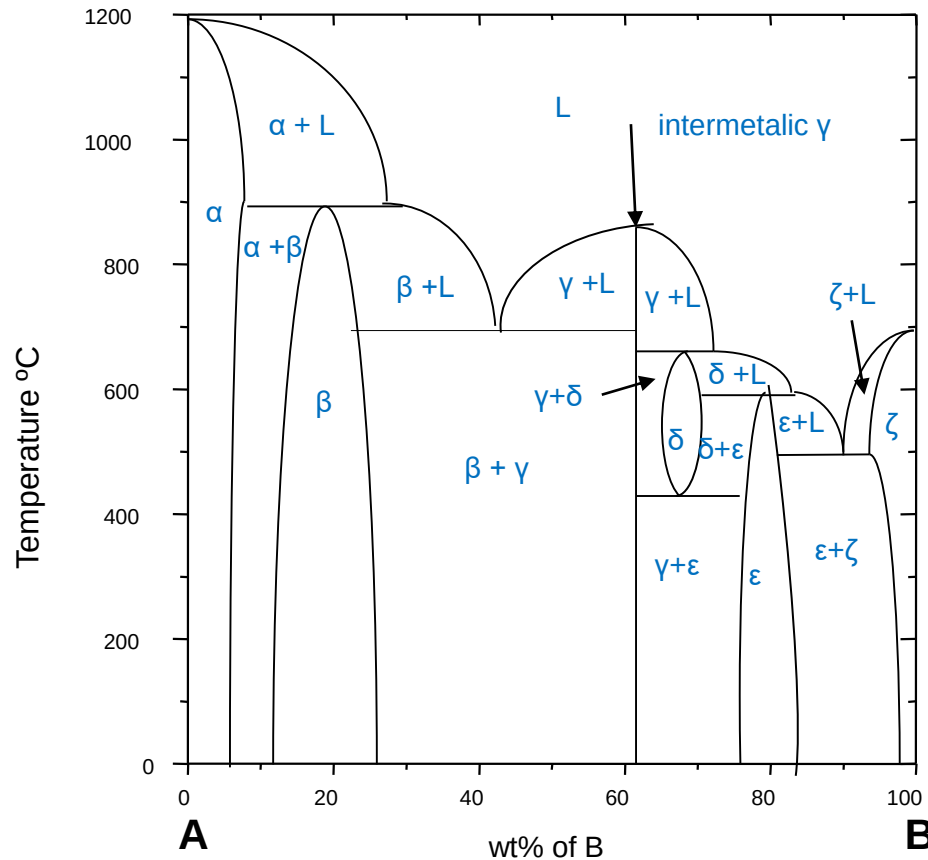


PROBLEM 9









β incongruent melting
 γ congruent melting
 δ incongruent melting
 ϵ incongruent melting